

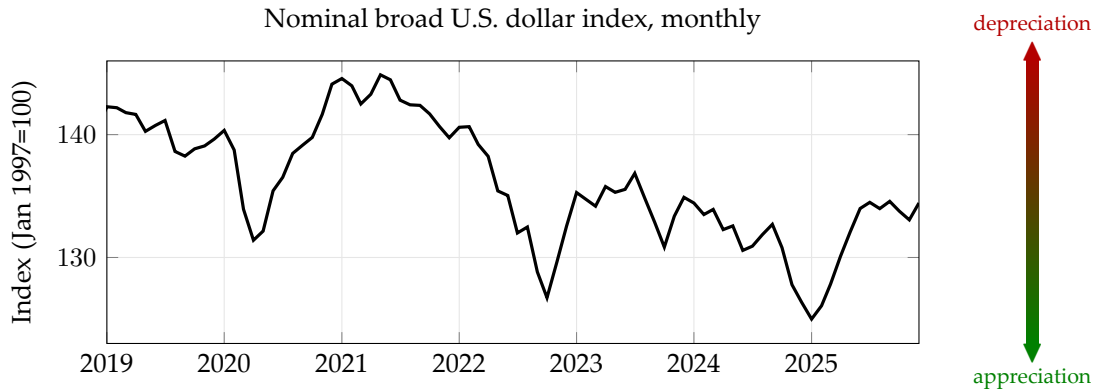
# ECON 1550: International Finance

## A Tour of the Class

# What is International Finance?

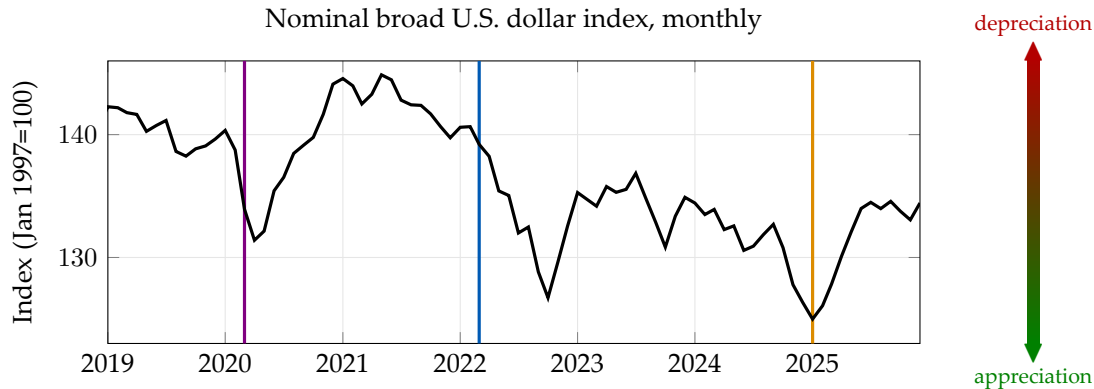
- Extension of Intermediate Macro to open economies
- Open economies:
  - Imports and Exports
  - $Y = C + I + G$  becomes  $Y = C + I + G + EX - IM$
- Why finance?
  - Exports and imports must be paid for!
  - If a country imports more than it exports, it must issue IOUs
  - IOUs are *financial assets*
  - $EX - IM$  = change in net foreign wealth

# Dollar Exchange Rate: 2019–present



Source: Board of Governors of the Federal Reserve System (US) via FRED (TWEXBMTH and TWEXBGSMTH, spliced).

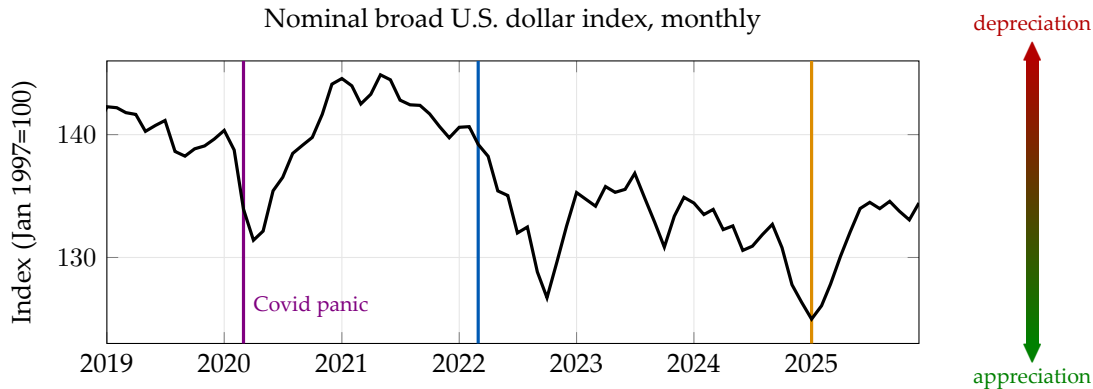
# Dollar Exchange Rate: 2019–present



Source: Board of Governors of the Federal Reserve System (US) via FRED (TWEXBMTH and TWEXBGSMTH, spliced).

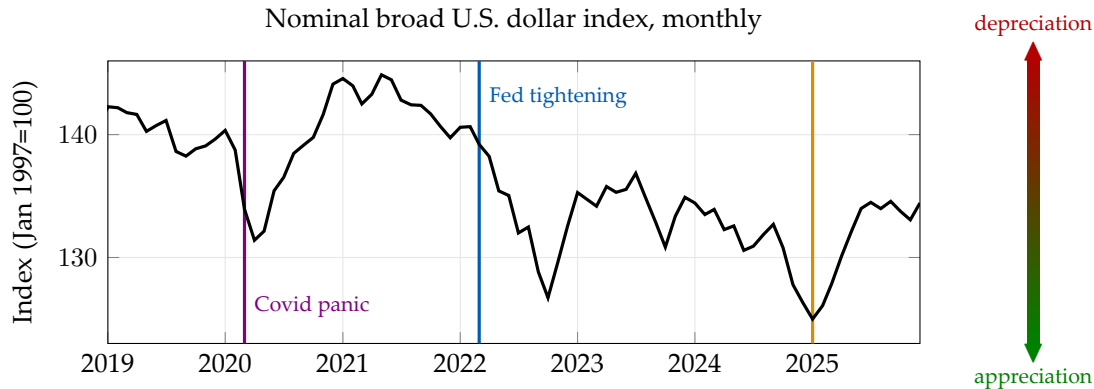


# Dollar Exchange Rate: 2019–present



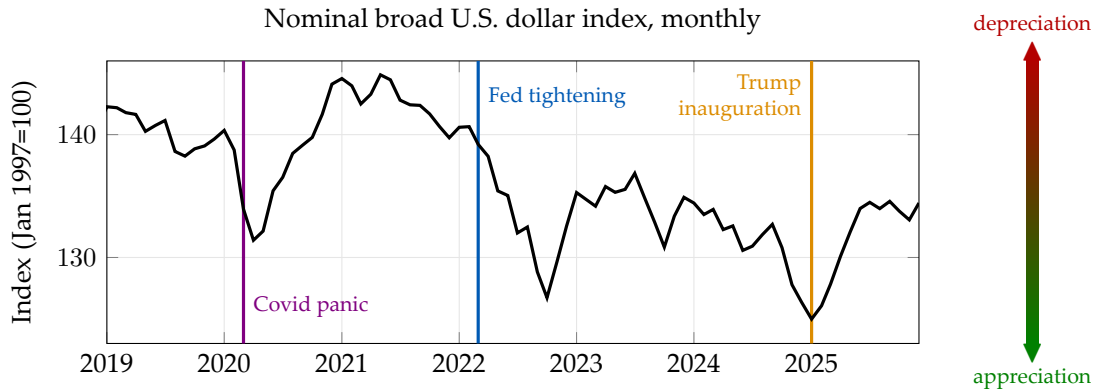
Source: Board of Governors of the Federal Reserve System (US) via FRED (TWEXBMTH and TWEXBGSMTH, spliced).

# Dollar Exchange Rate: 2019–present



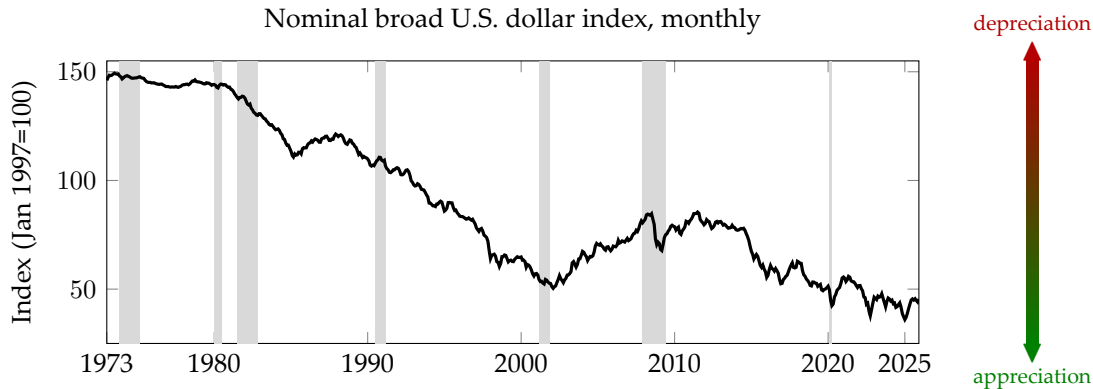
Source: Board of Governors of the Federal Reserve System (US) via FRED (TWEXBMTH and TWEXBGSMTH, spliced).

# Dollar Exchange Rate: 2019–present



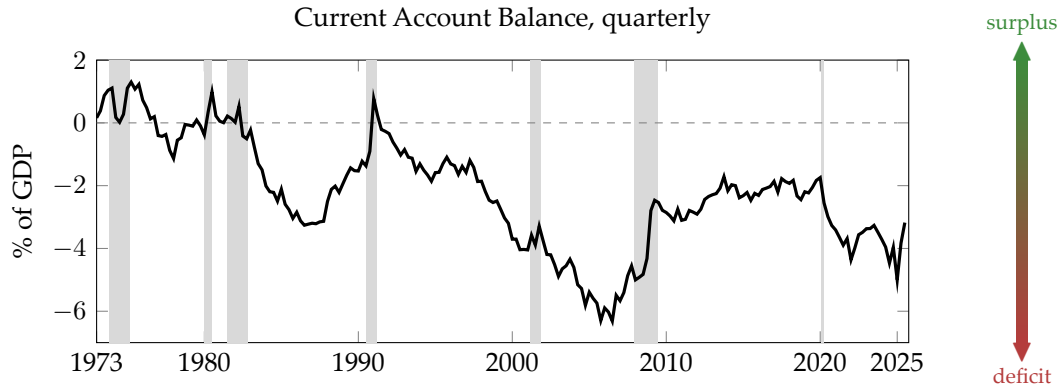
Source: Board of Governors of the Federal Reserve System (US) via FRED (TWEXBMTH and TWEXBGSMTH, spliced).

# Dollar Exchange Rate: 1973–present

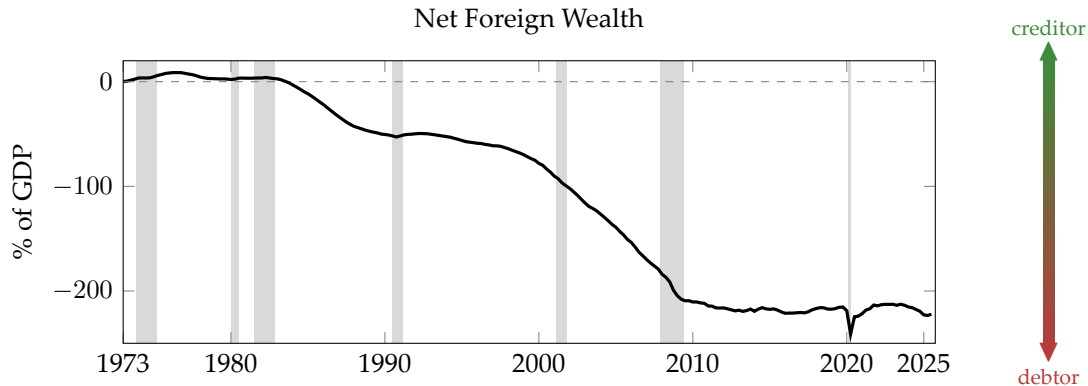


Source: Board of Governors of the Federal Reserve System (US) via FRED (TWEXBMTH and TWEXBGSMTH, spliced).

# U.S. Current Account Balance: 1973–present



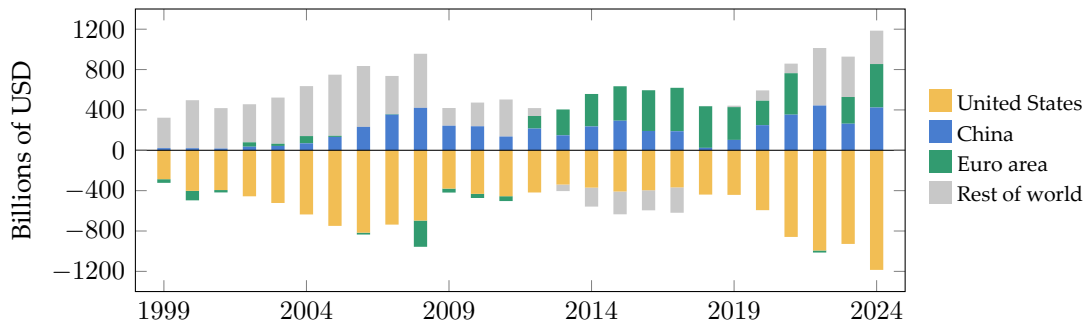
# Net Foreign Wealth



Source: BEA via FRED (NETFI, GDP). Constructed NFW = cumulative current account / GDP.

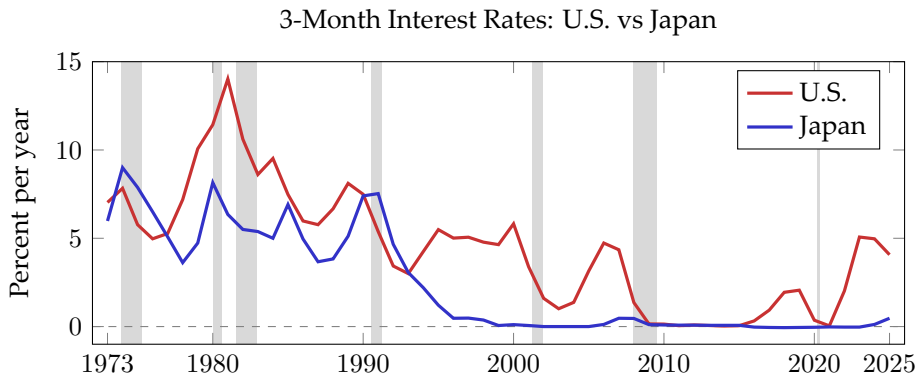
# Global External Imbalances

Current Account Balances by Region, 1999–2024



Source: IMF DataMapper API, WEO, indicator BCA (current account balance, billions USD). Rest of world = residual to balance global accounts.

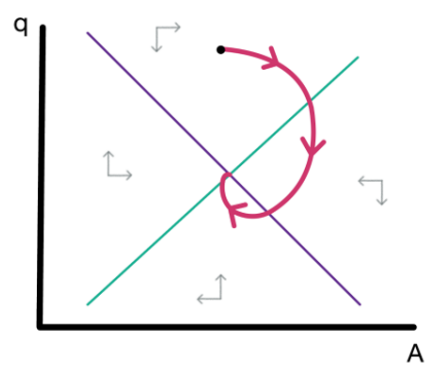
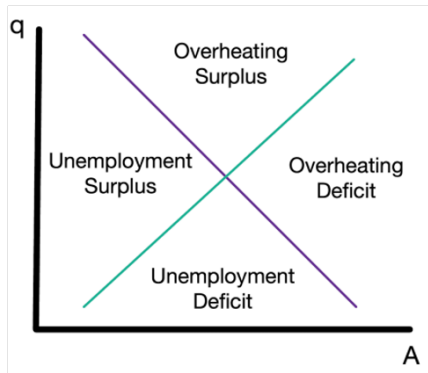
# Large Interest Rate Differentials



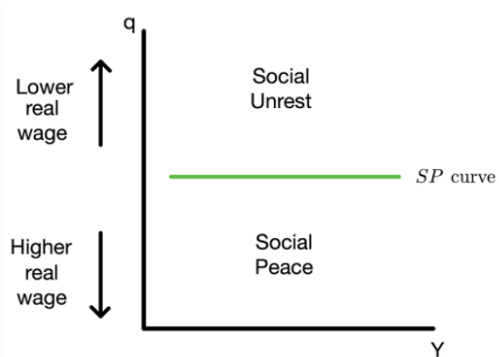
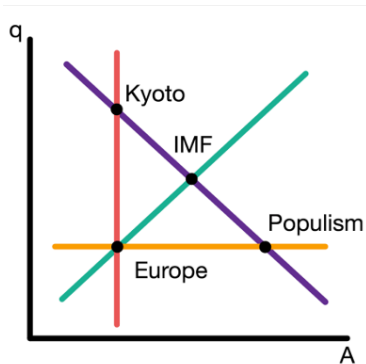
Source: FRED. US: TB3MS (3-month T-bill). Japan: IRSTCB01JPM156N (central bank rate, 1973–84), IRSTCI01JPM156N (call money rate, 1985–present). Annual averages.



# Exchange Rate Dynamics



# Policy Trade-offs



# Takeaways

- **To remember:** International finance connects exchange rates, trade balances, interest rates, and policy choices
- **To do for Friday:**
  - Read the note “Models in Economics”
  - Complete the “Getting Started” module on Canvas

ECON 1550: International Finance

Models in Economics

Bottom line to remember

“All models are wrong, some models are useful”

# Example: Uncovered Interest Parity (UIP)

$$R_{\$} = R_{¥} + \frac{E_{\$/¥}^e - E_{\$/¥}}{E_{\$/¥}}$$

- Does not fit the data well
- We can still use it to determine  $E_{\$/¥}$  given  $R_{\$}$ ,  $R_{¥}$ , and  $E_{\$/¥}^e$
- Combined with goods and money market equilibrium, we can understand (some or all) effects of monetary policy on  $E_{\$/¥}$

## Example: UIP (continued)

$$R_{\$} = R_{¥} + \frac{E_{\$/¥}^e - E_{\$/¥}}{E_{\$/¥}} + rp$$

- How much did we miss? Add an error term and call it **risk premium**
- If  $rp$  is independent of monetary policy, we did not miss anything
- If  $rp$  depends on monetary policy, we still captured one channel

# Types of Variables

## **Endogenous**

- Explained within the model

## **Exogenous**

- Taken as given
- Not explained by the model

## **Parameters**

- Exogenous; do not depend on policy



# Types of Equations

## **Identities**

- Hold by definition or construction

## **Behavioral**

- Capture behavior that we include in a model
- Hold by assumption

## **Equilibrium conditions**

- Supply equals demand
- Hold by “economic logic”

# Solving a Model, Solving for a Variable

- “Solving for a variable” means expressing that variable in terms of exogenous variables only
- “Solving a model” means solving for all endogenous variables

# Example 1

## Exogenous variables

| Variable | Description                       |
|----------|-----------------------------------|
| $T$      | taxes                             |
| $Y$      | income                            |
| $c_1$    | marginal propensity<br>to consume |

## Endogenous variables

| Variable | Description          | Equation           | Type of equation |
|----------|----------------------|--------------------|------------------|
| $C$      | consumption          | $C = c_1 Y_D$      | behavioral       |
| $Y_D$    | disposable<br>income | $Y_D \equiv Y - T$ | identity         |

**Solution  
and  
Intuition**

# Example 2

## Exogenous variables

| Variable | Description                    |
|----------|--------------------------------|
| $C$      | consumption                    |
| $Y$      | income                         |
| $c_1$    | marginal propensity to consume |

**Solution  
and  
Intuition**

## Endogenous variables

| Variable | Description       | Equation           | Type of equation |
|----------|-------------------|--------------------|------------------|
| $T$      | taxes             | $Y_D \equiv Y - T$ | identity         |
| $Y_D$    | disposable income | $C = c_1 Y_D$      | behavioral       |

# Example 2

## Exogenous variables

| Variable | Description                    |
|----------|--------------------------------|
| $C$      | consumption                    |
| $Y$      | income                         |
| $c_1$    | marginal propensity to consume |

## Endogenous variables

| Variable | Description       | Equation           | Type of equation |
|----------|-------------------|--------------------|------------------|
| $T$      | taxes             | $Y_D \equiv Y - T$ | identity         |
| $Y_D$    | disposable income | $C = c_1 Y_D$      | behavioral       |

**Solution  
and  
Intuition**

# Shocks

- Changes in exogenous variables
- Including changes in parameters
- Usually unforeseen, unforeseeable, or random

# ECON 1550: International Finance

IS-LM-PC in one day

# Announcements

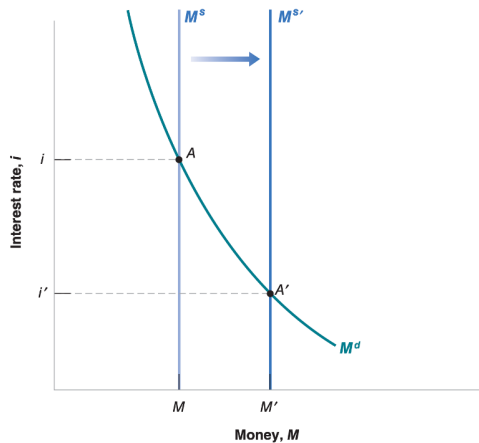
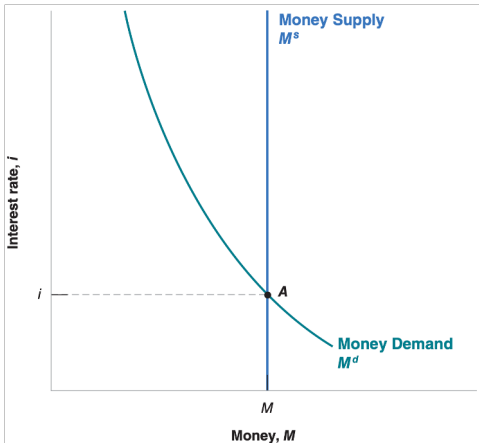
- Problem Set 1 due next Wednesday
- Read pages 13–23 of textbook before Friday lecture
- Office hours for this week on Canvas
- Section will be announced
- Review of Intermediate Macro note posted on Canvas



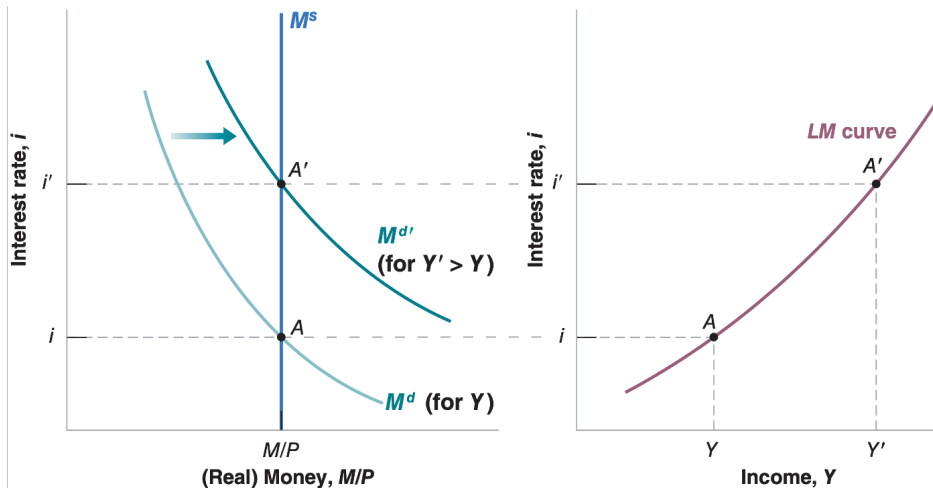
# Agenda

- Close loop on Tour of the class and models
- IS-LM
- IS-LM-PC

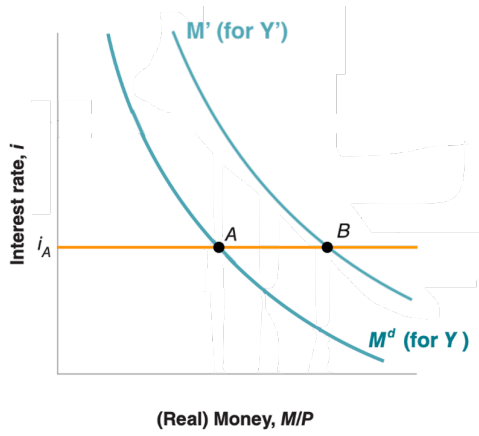
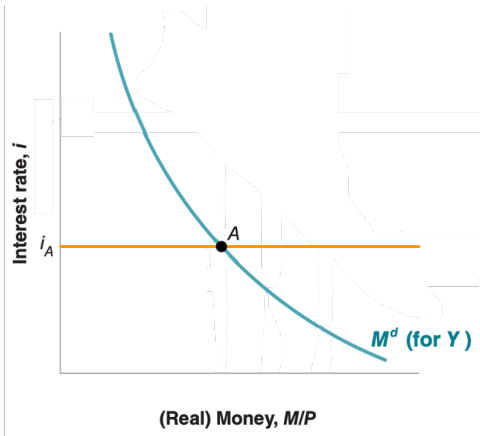
# Exogenous money supply...



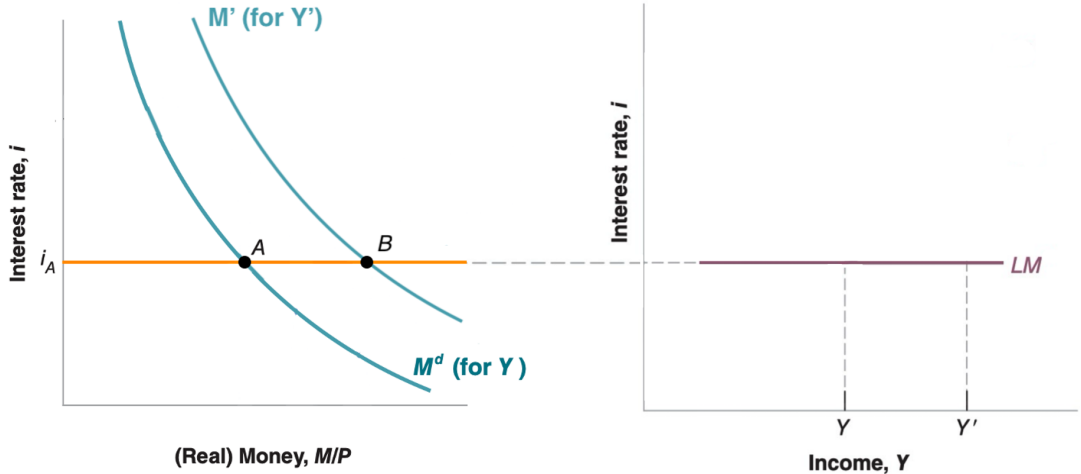
...leads to an upward sloping LM



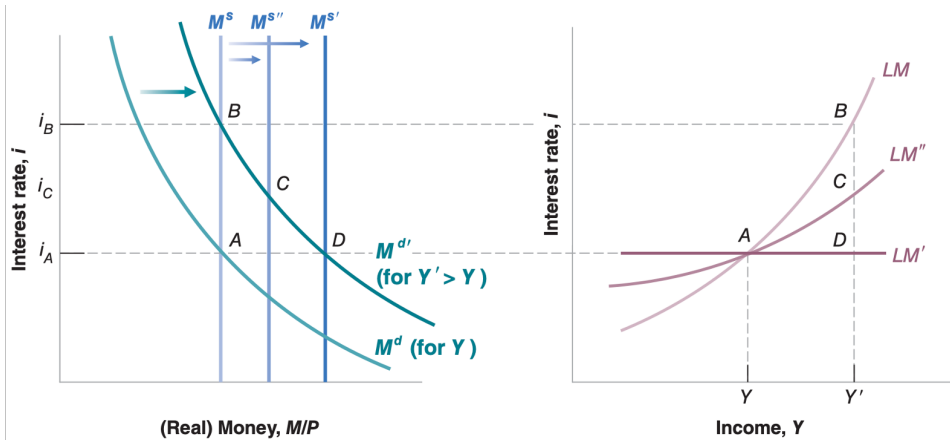
# Exogenous interest rates...



...lead to a flat LM



# Endogenous Money Supply



# A Model for the Labor Market

(Wage setting)

$$W = P^e F(u, z)$$

(Price setting)

$$P = (1 + m)W$$

(Definition of expected inflation)

$$\pi^e = P^e / P - 1$$

(Linear function F)

$$F(u, z) = 1 - \alpha u + z$$

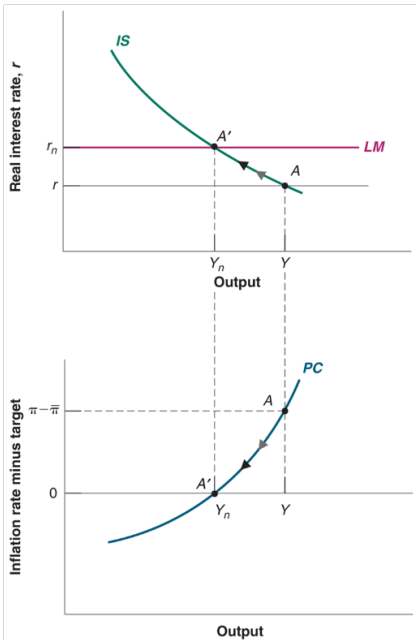
# Phillips Curve

- Three equivalent forms

$$\pi = \pi^e - \alpha(u - u^n)$$

$$\pi = \pi^e + \alpha(Y - Y^n)$$

$$\pi = \pi^e + (m + z) - \alpha u$$





ECON 1550: International Finance

National Income Accounting for  
Open Economies

# Announcements

- Problem Set 1 due today before midnight
- Solutions and Problem Set 2 posted right after
- Read pages 58–67 of textbook before Wednesday lecture
- Fill in survey for office hours and section (right now!)

# Office Hours and Section Availability

Please fill in your availability:



<https://www.when2meet.com/?34820140-APJfJ>

# Agenda

- Balance of payments account
- Exchange rate determination with asset approach

# Macro Review: GDP

**Gross Domestic Product (GDP)** measures the total value of:

- **Production:** All final goods and services produced within a country
- **Income:** All income earned from production within a country
- **Value added:** Sum of value added at each stage of production

All three approaches yield the same number.

# Closed Economy

$$Y = C + I + G$$

- All output is either consumed, invested, or purchased by government
- No trade with the rest of the world

# Open Economy with GDP

$$GDP = C + I + G + \underbrace{EX - IM}_{NX}$$

- $EX$  = Exports (domestic goods sold abroad)
- $IM$  = Imports (foreign goods purchased domestically)
- $NX$  = Net exports (trade balance)

# GDP and GNP

- **GDP:** Value of production *within a country's borders*
  - Regardless of who owns the factors of production
- **GNP:** Value of production by *a country's residents*
  - Regardless of where production takes place
- **Relationship:**

$$\text{GNP} = \text{GDP} + \text{Net Income from Abroad}$$

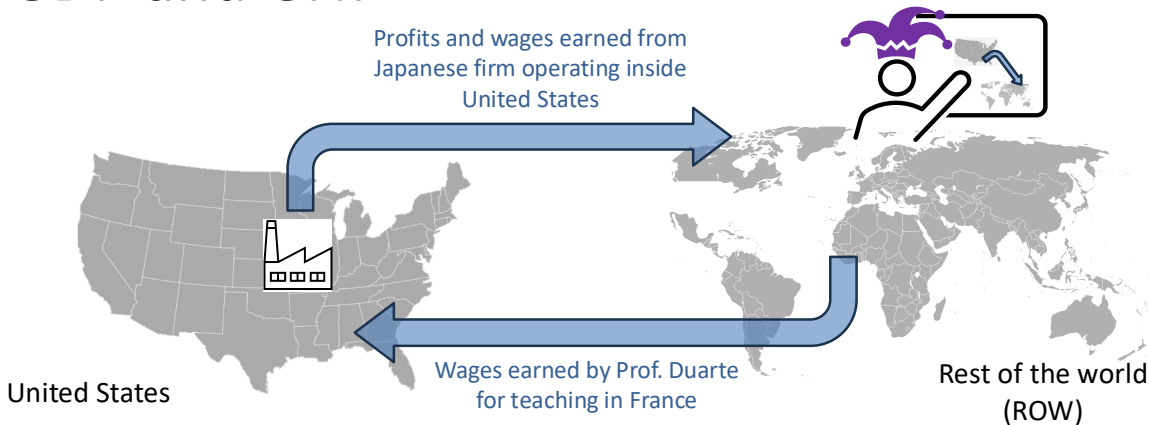


# Open Economy with GNP

$$GNP = C + I + G + CA$$

- $CA$  = Current account
- $CA = NX +$  Net income from abroad

# GDP and GNP



Net Income = wages from Prof. Duarte teaching in France — profits and wages from firm operating inside US

United States GNP = United States GDP + Net Income

# National Saving: Closed Economy

**National saving** = output not used for consumption or government spending

$$\begin{aligned} S &\equiv Y - C - G \\ &= (C + I + G) - C - G \\ &= I \end{aligned}$$

**In a closed economy:**  $S = I$

# National Saving: Open Economy

Starting from  $Y = C + I + G + NX$ :

$$S \equiv Y - C - G$$

$$= I + NX$$

**In an open economy:**  $S = I + NX$

# National Saving: Open Economy

Rearranging:  $S - I = NX$

- If  $S > I$ : Trade surplus, country is a net lender
- If  $S < I$ : Trade deficit, country is a net borrower

# Private Saving

**Private saving** = disposable income that is saved, not consumed

$$S^p \equiv Y - T - C$$

# Public (Government) Saving

**Government saving** = tax revenue minus government spending

$$S^g \equiv T - G$$

- If  $T > G$ : Budget surplus ( $S^g > 0$ )
- If  $T < G$ : Budget deficit ( $S^g < 0$ )

# Total Saving

**Total national saving:**

$$\begin{aligned} S &= S^p + S^g \\ &= (Y - T - C) + (T - G) \\ &= Y - C - G \end{aligned}$$

Using  $S = I + NX$  from before, we get:

$$S^p + S^g = I + NX$$



# Balance of Payments

|                          |      |
|--------------------------|------|
| <b>Current Account</b>   | −200 |
| <b>Capital Account</b>   | −1   |
| <b>Financial Account</b> | −201 |

$$\begin{array}{rclcl} \text{Current} & & \text{Capital} & & \text{Financial} \\ \text{Account} & + & \text{Account} & = & \text{Account} \\ (-200) & + & (-1) & = & -201 \end{array}$$

# Balance of Payments

|                                |      |
|--------------------------------|------|
| <b>Current Account</b>         | -200 |
| <b>Capital Account</b>         | -1   |
| <b>Financial Account</b>       | -201 |
| <b>Statistical discrepancy</b> | 0    |

$$\begin{array}{rclcl} \text{Statistical} & & \text{Financial} & & \\ \text{Discrepancy} & = & \text{Account} & - & \left( \begin{array}{cc} \text{Current} & \text{Capital} \\ \text{Account} & \text{Account} \end{array} \right) \\ 0 & = & -201 & - & \left[ \begin{array}{cc} (-200) & + & (-1) \end{array} \right] \end{array}$$

## U.S. Balance of Payments Accounts for 2025-Q3 (\$ bn)

|                 |      |
|-----------------|------|
| Current Account | -226 |
|-----------------|------|

|                 |    |
|-----------------|----|
| Capital Account | -1 |
|-----------------|----|

|                   |      |
|-------------------|------|
| Financial Account | -410 |
|-------------------|------|

|                         |      |
|-------------------------|------|
| Statistical discrepancy | -183 |
|-------------------------|------|

$$\begin{array}{rclcl} \text{Statistical} & & \text{Financial} & & \\ \text{Discrepancy} & = & \text{Account} & - & \left( \begin{array}{cc} \text{Current} & \text{Capital} \\ \text{Account} & \text{Account} \end{array} \right) \\ -183 & = & -410 & - & [ (-226) + (-1) ] \end{array}$$

Source: BEA

---

## Current Account

- 1 Exports total
- 2     Goods
- 3     Services
- 4     Income receipts (primary income)
- 5 Imports total
- 6     Goods
- 7     Services
- 8     Income payments (primary income)
- 9 Net unilateral transfers (secondary income)
- 10 **Balance on current account**

## Capital Account

- 11 Balance on capital account
-

---

## Financial Account

12 Net U.S. acquisition of foreign financial assets

13     Official reserve assets

14     Other assets

15 Net U.S. incurrence of domestic liabilities

16     Official reserve assets

17     Other liabilities

18 Financial derivatives, net

19 **Net financial flows**

---

20 **Statistical discrepancy**

---

## Bottom line to remember

Whenever goods cross borders, assets move  
in the opposite direction.

ECON 1550: International Finance

# Exchange Rates and the Foreign Exchange Market: An Asset Approach

# A model of exchange rate determination

## Exogenous variables

| Variable | Description            |
|----------|------------------------|
| $R$      | Domestic interest rate |
| $R^*$    | Foreign interest rate  |
| $E^e$    | Expected exchange rate |

## Endogenous variables

| Variable | Description   | Equation                      | Type of equation      |
|----------|---------------|-------------------------------|-----------------------|
| $E$      | Exchange rate | $R = R^* + \frac{E^e - E}{E}$ | Equilibrium condition |



# A model of exchange rate determination

- Two investment opportunities
  - Domestic bond with return  $R_{\$}$  in Dollars
  - Foreign bond with return  $R^*$  in Euros
- To be indifferent between the two investments, the **uncovered interest parity condition** must hold:

$$(\text{UIP}): R_{\$} = R^* + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

# Exchange Rates

# Bonds

# Realized vs Expected Returns

# Expected Returns of Foreign Bond

# Approximations

# Uncovered Interest Parity

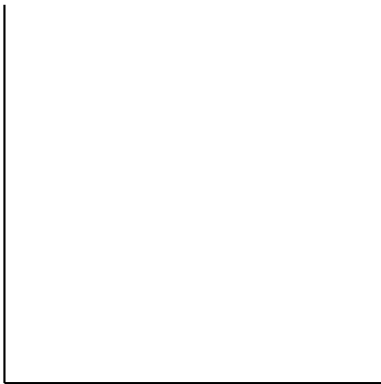
- For both strategies to have the same expected return

$$1 + R_{\$} = \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} (1 + R^*)$$

- Approximating

$$R_{\$} = R^* + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

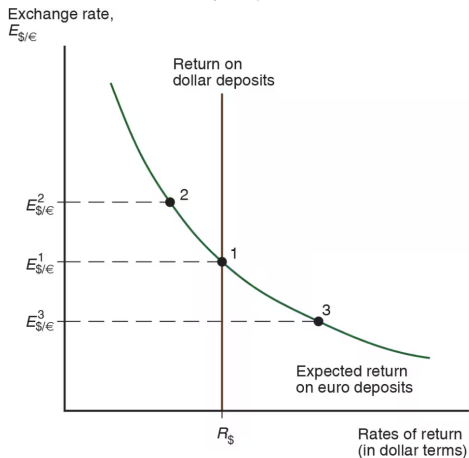
# Equilibrium in Foreign Exchange Market



$$\text{UIP: } R_{\$} = R^{*} + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

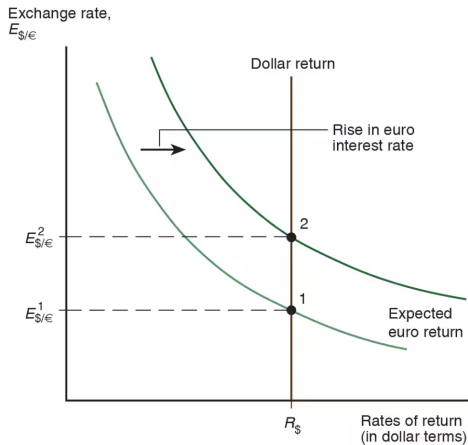


# Equilibrium in Foreign Exchange Market



$$\text{UIP: } R_{\$} = R^{*} + \frac{E_{\$/\text{EUR}}^e}{E_{\$/\text{EUR}}} - 1$$

# Shocks: Rise in Euro Interest Rate



# The carry trade

- Borrowing at “low” rate  $R$  and lending at “high” rate  $R^*$  is a **carry trade**

$$\begin{array}{l} \text{expected return} \\ \text{on carry trade} \end{array} = R^* + \left( \frac{E^e}{E} - 1 \right) - R + \text{risk premium}$$

- Risk: Future exchange rate is not known when we start the carry trade,  $E^e$  can be different from the realized future exchange rate

# Empirical Failure of UIP

Eight currency portfolios sorted by interest rate differential (portfolio 1 = lowest rates, portfolio 8 = highest).

High interest rate currencies earn higher mean excess returns and have higher Sharpe ratios.

Annual data, 1953–2002, US investor perspective.

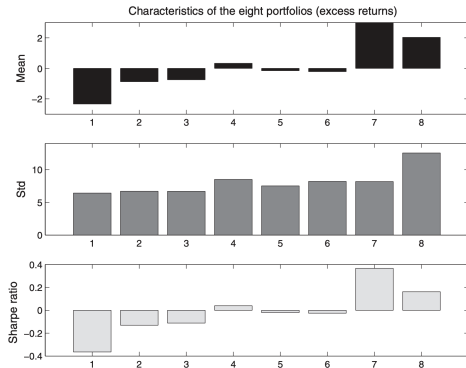


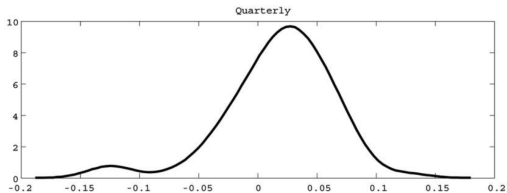
FIGURE 1. EIGHT CURRENCY PORTFOLIOS

Source: [Lustig & Verdelhan \(2007\)](#), *American Economic Review*

# Explanations For Carry Risk Premium

- Crash risk / peso problem
- Correlation with consumption
- Global Volatility Risk
- Liquidity
- Constrained intermediaries

# Crash Risk



Kernel density of excess returns on a carry trade portfolio (long three high interest currencies, short three low interest currencies). Right: USD/JPY exchange rate, 1996–2000.



Fig. 1. U.S. dollar/Japanese yen exchange rate from 1996 to 2000

Source: Brunnermeier, Nagel, and Pedersen (2008), *NBER Macroeconomics Annual*

# Consumption Risk

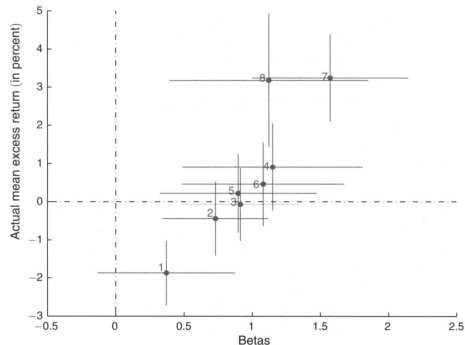
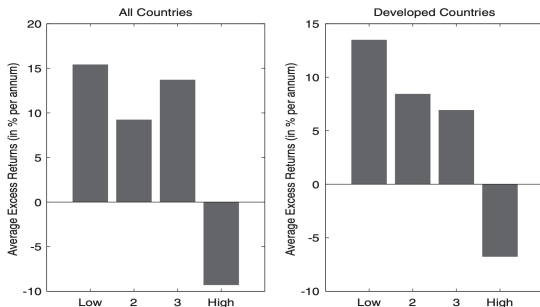


FIGURE 3. DURABLE CONSUMPTION GROWTH BETAS AND AVERAGE CURRENCY EXCESS RETURNS

The dots represent point estimates; the lines represent one standard deviation above and below. The sample is 1953–2008, annual data. Higher consumption betas correspond to higher currency excess returns.

Source: Lustig & Verdelhan (2011), *American Economic Review*

# Global Volatility Risk



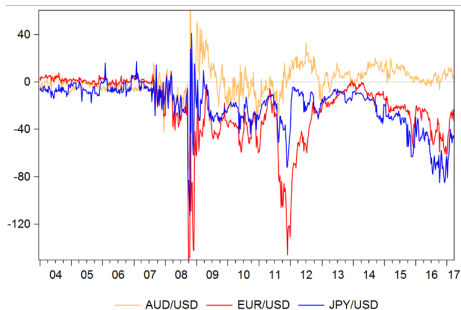
**Figure 2. Excess returns and volatility.** The figure shows mean excess returns for carry trade portfolios conditional on global FX volatility innovations being within the lowest to highest quartile of its sample distribution (four categories from “Low” to “High” shown on the x-axis of each panel). The bars show average excess returns for being long in portfolio 5 (largest forward discounts) and short in portfolio 1 (lowest forward discounts). The left panel shows results for all countries, while the right panel shows results for developed countries. The sample period is November 1983 to August 2009.

Mean excess returns for carry trade portfolios conditional on global FX volatility innovations being within the lowest to highest quartile. Carry trade earns high returns when volatility is low, but suffers large losses when volatility is high.

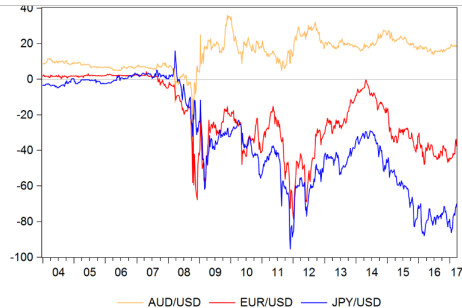
Source: Menkhoff, Sarno, Schmeling, and Schrimpf (2012), *Journal of Finance*



# Violations of *Covered* Interest Parity



3-month basis:  $b_{3m}$

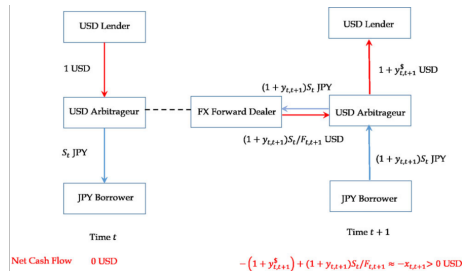
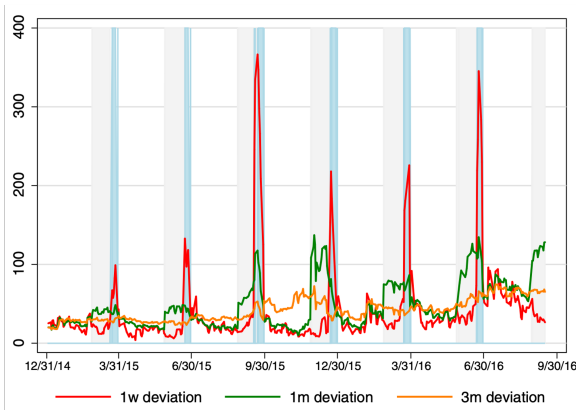


3-year basis:  $b_{3y}$

Source: Sushko, Borio, McCauley, and McGuire, "The Failure of Covered Interest Parity"

# Intermediaries are Constrained

Illustration of quarter-end dynamics of CIP deviations.



Source: Du, Tepper, and Verdelhan, "Deviations from Covered Interest Rate Parity," *Journal of Finance*

ECON 1550: International Finance

# Money, Interest Rates, and Exchange Rates

# A model of the money market

| Exogenous variables |              | Endogenous variables |                        |                   |                       |
|---------------------|--------------|----------------------|------------------------|-------------------|-----------------------|
| Variable            | Description  | Variable             | Description            | Equation          | Type of equation      |
| $Y$                 | Real income  | $R$                  | Domestic interest rate | $M^d/P = L(R, Y)$ | Behavioral equation   |
| $M^s$               | Money supply | $M^d$                | Money demand           | $M^d = M^s$       | Equilibrium condition |
| $P$                 | Price level  |                      |                        |                   |                       |

# Description of the Money Market

- There are only two assets: money and domestic bonds
- Money
  - Can be used for transactions
  - Pays no interest
- Bonds
  - Cannot be used for transactions
  - Pay interest  $i \geq 0$

# Money Demand

- Money demand is higher when:
  - Higher price level  $P \rightarrow$  need more money to buy goods
  - Lower nominal interest rate  $R \rightarrow$  bonds less attractive
  - Higher income  $Y \rightarrow$  want more money to buy more
- Capture idea with a behavioral equation

$$M^d = P \times L(\underset{(-)}{R}, \underset{(+)}{Y})$$

# Real money demand

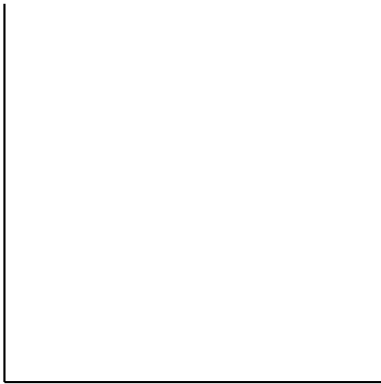
- Convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

and real money supply

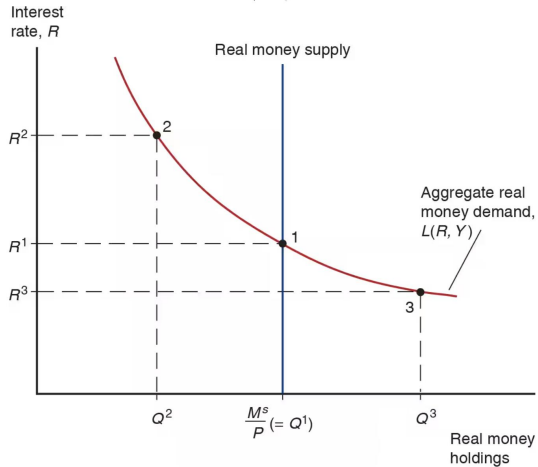
$$\frac{M^s}{P}$$

# Equilibrium in money market





# Equilibrium in money market



# Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

## Many types of money 1/2

- Commodity money: a physical commodity (like gold) that is used as money
- Convertible paper money: a piece of paper that can be exchanged by a commodity

## Many types of money 2/2

- Fiat money: issued by a central bank, not backed by any commodity
- Digital currency: not backed by any commodity, privately issued, electronic payments

# Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

| Assets  | Liabilities | Assets       | Liabilities | Assets    | Liabilities |
|---------|-------------|--------------|-------------|-----------|-------------|
| \$0     | \$0         | \$1 currency | \$1 money   | \$1 bonds | \$1 money   |
| $t = 0$ |             | $t = 1$      |             | $t = 2$   |             |

# Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves

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# Money, Interest Rates, and Exchange Rates

# A model of the money market

| Exogenous variables |              | Endogenous variables |                        |                   |                       |
|---------------------|--------------|----------------------|------------------------|-------------------|-----------------------|
| Variable            | Description  | Variable             | Description            | Equation          | Type of equation      |
| $Y$                 | Real income  | $R$                  | Domestic interest rate | $M^d/P = L(R, Y)$ | Behavioral equation   |
| $M^s$               | Money supply | $M^d$                | Money demand           | $M^d = M^s$       | Equilibrium condition |
| $P$                 | Price level  |                      |                        |                   |                       |



# Description of the Money Market

- There are only two assets: money and domestic bonds
- Money
  - Can be used for transactions
  - Pays no interest
- Bonds
  - Cannot be used for transactions
  - Pay interest  $R \geq 0$

# Money Demand

- Money demand is higher when:
  - Higher price level  $P \rightarrow$  need more money to buy goods
  - Lower nominal interest rate  $R \rightarrow$  bonds less attractive
  - Higher income  $Y \rightarrow$  want more money to buy more
- Capture idea with a behavioral equation

$$M^d = P \times L(\underset{(-)}{R}, \underset{(+)}{Y})$$

# Real money demand

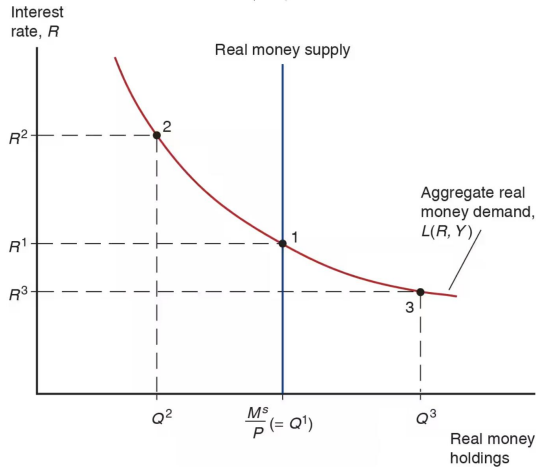
- Convenient to write in terms of real money demand

$$\frac{M^d}{P} = L(R, Y)$$

and real money supply

$$\frac{M^s}{P}$$

# Equilibrium in money market



# Money is defined by its function

- Medium of exchange
- Store of value
- Unit of account

## Many types of money (1/2)

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- Convertible paper money: a piece of paper that can be exchanged for a commodity

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# Fiat money

1. Central bank liabilities are special because they are the unit of account.
2. Currency is a promise to deliver future central bank liabilities.

| Assets  | Liabilities | Assets       | Liabilities | Assets    | Liabilities |
|---------|-------------|--------------|-------------|-----------|-------------|
| \$0     | \$0         | \$1 currency | \$1 money   | \$1 bonds | \$1 money   |
| $t = 0$ |             | $t = 1$      |             | $t = 2$   |             |



# Monetary policy

- Deposits at the Fed are called reserves
- “The” interest rate is just interest on reserves

# Short-run FX and money market model

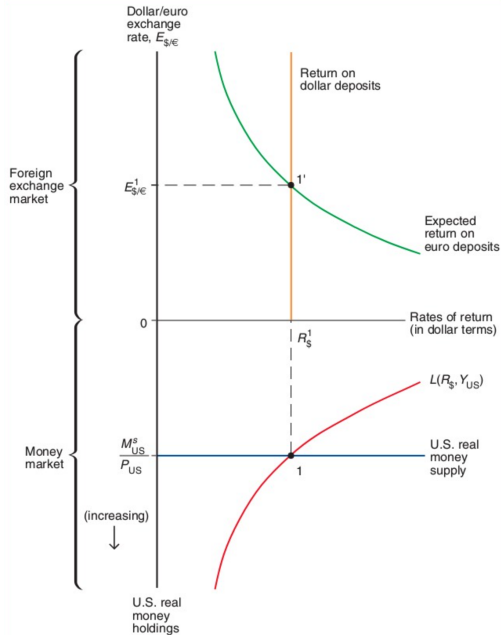
## Exogenous variables

| Variable | Description            |
|----------|------------------------|
| $R^*$    | Foreign interest rate  |
| $E^e$    | Expected exchange rate |
| $Y$      | Real income            |
| $M^s$    | Money supply           |
| $P$      | Price level            |

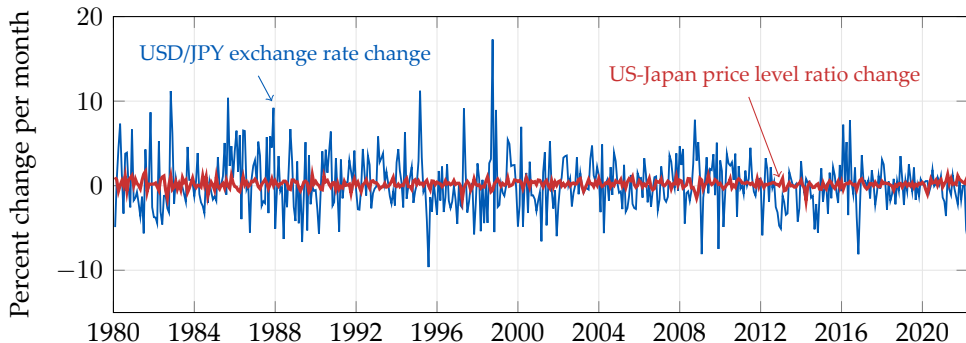
## Endogenous variables

| Variable | Description            | Equation                      | Type of equation      |
|----------|------------------------|-------------------------------|-----------------------|
| $E$      | Exchange rate          | $R = R^* + \frac{E^e - E}{E}$ | Equilibrium condition |
| $R$      | Domestic interest rate | $M^d/P = L(R, Y)$             | Behavioral equation   |
| $M^d$    | Money demand           | $M^d = M^s$                   | Equilibrium condition |

# Short-run equilibrium in FX and money markets



# Exchange rates are very volatile



Source: [FRED/CPIAUCSL](#); [FRED/CPALCY01JPM661N](#); [FRED/DEXJPUS](#) (end-of-month, inverted to USD/JPY).

# Conceptual definition of long-run equilibrium

- Hypothetical equilibrium that would result if economy runs indefinitely with no shocks
- Equivalently, the hypothetical equilibrium that would occur if prices were perfectly flexible and always adjusted instantaneously and frictionlessly

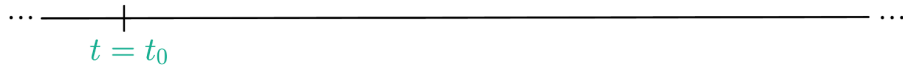
# Definition of the long run in the model

1. In the long run,  $E = E^e$ 
  - By definition of the long run, all variables constant
  - By assumption of rational expectations, expectations must be correct in the long run
  - In the model, it means  $E_{LR} = E_{LR}^e$

# Assumptions about the short run

1. In the short run,  $P$  is fixed
  - Because prices are “sticky”
  - In the model, it means  $P_0 = P_{SR}$
2. In the short run,  $E^e$  equals long-run  $E$ 
  - By assumption of rational expectations
  - In the model, it means  $E_{SR}^e = E_{LR}$

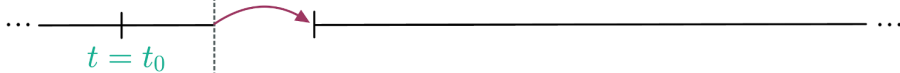
Initial long run  
equilibrium



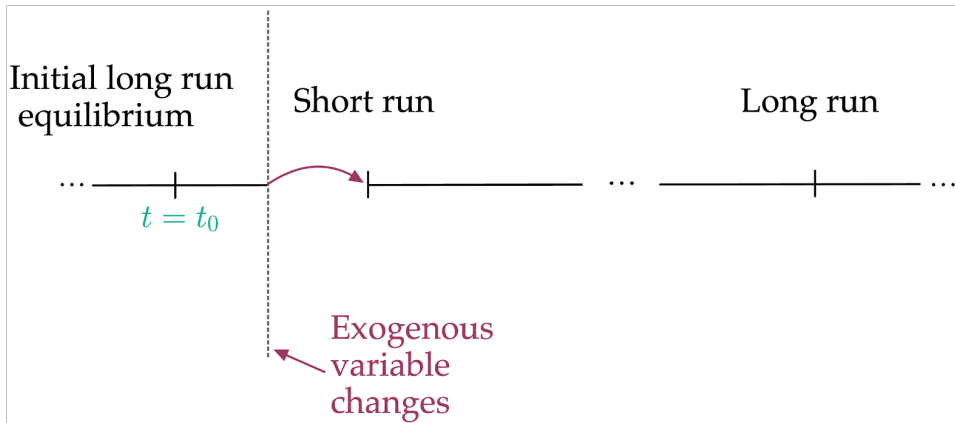


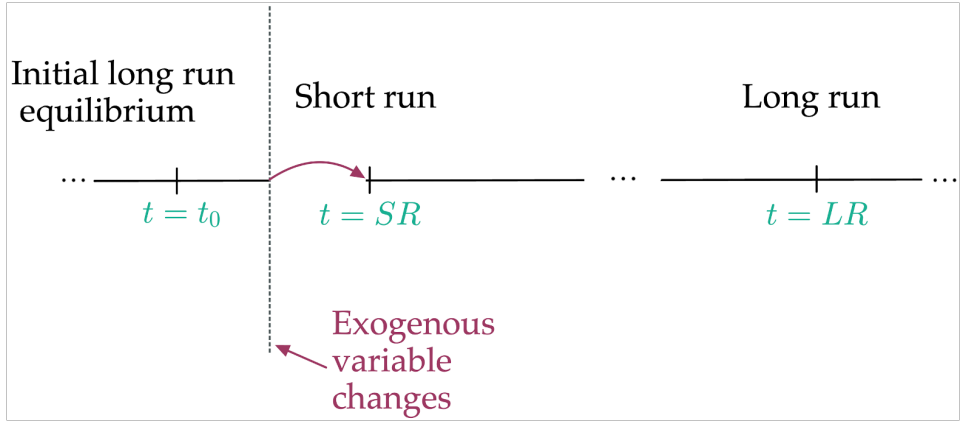
Initial long run  
equilibrium

Short run



Exogenous  
variable  
changes





# Results in the long run (1/3)

1. Long-run interest rate  $R_{LR}$  from FX market
  - Using  $E_{LR} = E_{LR}^e$  and UIP:

$$R_{LR} = R^* + \frac{E_{LR}^e}{E_{LR}} - 1$$

$$\Rightarrow R_{LR} = R^*$$

## Results in the long run (2/3)

### 2. Long-run price level $P_{LR}$ from the money market

- Using  $R_{LR} = R^*$  and money market equilibrium condition:

$$\frac{M^s}{P_{LR}} = L(R_{LR}, Y) = L(R^*, Y)$$

$$\Rightarrow P_{LR} = \frac{M^s}{L(R^*, Y)}$$

## Results in the long run (3/3)

### 3. Long-run exchange rate $E_{LR}$ from PPP (purchasing power parity)

- Because  $E_{LR}$  is a nominal price, in the long run it moves proportionally to the price level:

$$E_{LR} \propto P_{LR}$$

# Results in the short run (1/2)

1. Short-run exchange rate  $E_{SR}$  from UIP with  $E^e = E_{LR}$ 
  - Using UIP and the short-run assumptions:

$$R = R^* + \frac{E_{LR} - E_{SR}}{E_{SR}} \quad \Rightarrow \quad E_{SR} = \frac{E_{LR}}{1 + R - R^*}$$

## Results in the short run (2/2)

### 2. Short-run domestic interest rate $R_{SR}$ from the money market

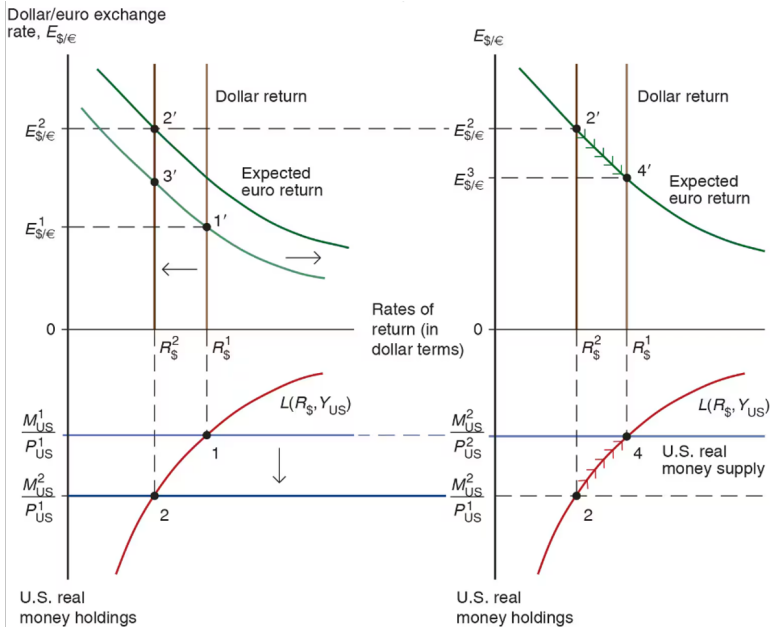
- Using money market equilibrium condition and  $P$  fixed at  $P_0$  in the short run:

$$\frac{M^s}{P_{SR}} = L(R_{SR}, Y) \quad \Rightarrow \quad \frac{M^s}{P_0} = L(R_{SR}, Y),$$

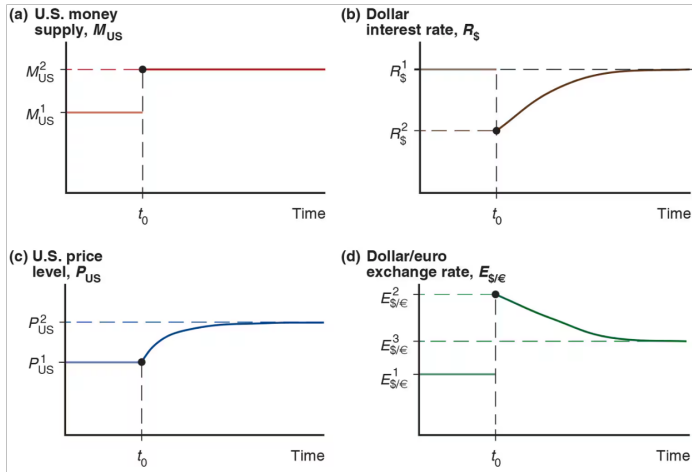
which can be solved for  $R_{SR}$



# Permanent Money Supply Changes



# Exchange Rate Overshooting



# Long-run FX and money market model

## Exogenous variables

| Variable | Description           |
|----------|-----------------------|
| $Y^n$    | Potential output      |
| $M^s$    | Money supply          |
| $R^*$    | Foreign interest rate |

## Endogenous variables

| Variable | Description            | Equation                      | Type of eq    |
|----------|------------------------|-------------------------------|---------------|
| $E$      | Exchange rate          | $R = R^* + \frac{E^e - E}{E}$ | Eq. condition |
| $M^d$    | Money demand           | $M^d/P = M^s/P$               | Eq. condition |
| $E^e$    | Expected exchange rate | $E^e = E$                     | Behavioral    |
| $R$      | Domestic interest rate | $P = \frac{M^d}{L(R, Y^n)}$   | Eq. condition |
| $P$      | Price level            | P fixed in the short run      | Behavioral    |

# Overshooting

1. Properties of initial, SR, LR equilibria
2. Dynamic FX and money market model
3. Example
  - Solve with equations
  - Solve graphically

# Properties of initial, SR, and LR equilibria

Long run:

- $E^e = E$

$$\Rightarrow E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR}$$

- $E$  proportional to  $P$  and  $M^s$

$$\Rightarrow E_0 = kM_0^s \quad \text{and} \quad E_{LR} = kM_{LR}^s \quad (k \text{ is a parameter})$$

# Properties of initial, SR, and LR equilibria

Short run:

- Prices are fixed

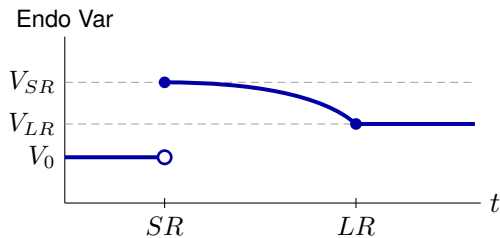
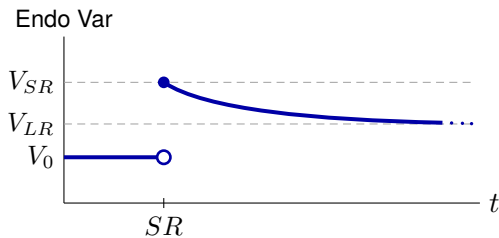
$$\Rightarrow P_{SR} = P_0$$

- Rational expectations

$$\Rightarrow E_{SR}^e = E_{LR}$$

# Transition between SR and LR

- Slow monotonic transition between SR value and LR value
- No indication of shape



# Dynamic FX and money market model

## Exogenous variables

| Variable | Description                                         |
|----------|-----------------------------------------------------|
| $M^s$    | Path of money supply                                |
| $R^*$    | Path of foreign interest rate                       |
| $Y$      | Path of income (GNP)                                |
| $k$      | Long-run proportionality constant for $E$ and $M^s$ |

## Endogenous variables

| Variable | Description            |
|----------|------------------------|
| $E$      | Exchange rate          |
| $E^e$    | Expected exchange rate |
| $P$      | Price level            |
| $R$      | Domestic interest rate |
| $M^d$    | Money demand           |



# Equations that apply at all times

$$\text{(UIP)} : R = R^* + \frac{E^e}{E} - 1 \quad \text{(equilibrium condition)}$$

$$\text{(MD)} : \frac{M^d}{P} = L\left(R, \underset{(-)}{Y}\right) \quad \text{(behavioral)}$$

$$\text{(MS = MD)} : \frac{M^s}{P} = \frac{M^d}{P} \quad \text{(equilibrium condition)}$$

# Equations that apply in SR equilibria only

(Sticky P) :  $P_{SR} = P_0$

(behavioral)

(RE) :  $E_{SR}^e = E_{LR}$

(behavioral)

# Equations that apply in LR equilibria only

$$\text{(PPP) or (Flex P)} : E_0 = kM_0^s, \quad E_{LR} = kM_{LR}^s \quad \text{(behavioral)}$$

$$\text{(Defn LR) or (RE)} : E_0^e = E_0, \quad E_{LR}^e = E_{LR} \quad \text{(behavioral)}$$

# Example

Need to specify:

1. Function  $L(R, Y)$
2. Path of exogenous variables

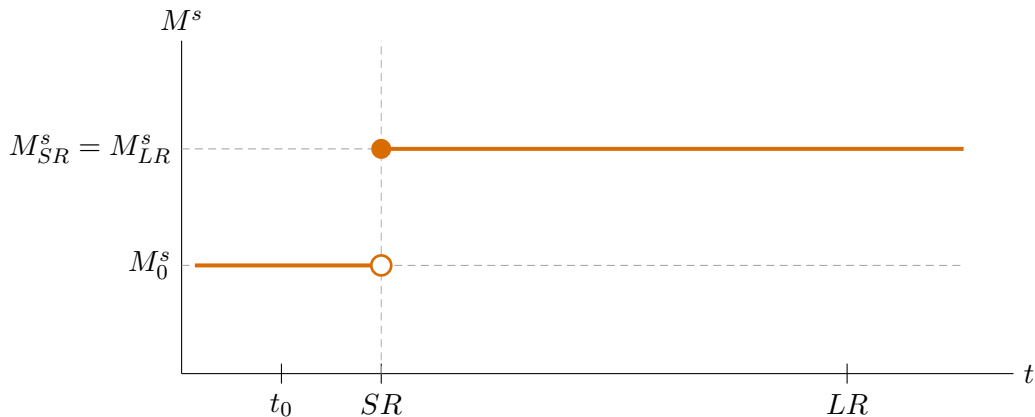
Assume:

1.  $L(R, Y) = \frac{Y}{1 + R}$

2.  $R^* = 0.08, Y = 1, k = 1$

$$\{M_0^s, M_{SR}^s, M_{LR}^s\} = \{1, 1.05, 1.05\}$$

# Example: Permanent increase in $M^s$



# Solution steps

1. Solve initial LR equilibrium ( $t = t_0$ )
2. Solve LR equilibrium ( $t = LR$ )
3. Solve SR equilibrium ( $t = SR$ )

# Solution sub-steps

## LR (steps 1 and 2)

- (a) Flex P / PPP  $\longrightarrow E$
- (b) Defn LR / RE  $\longrightarrow E^e$
- (c) UIP  $\longrightarrow R$
- (d) MS = MD  $\longrightarrow P$

## SR (step 3)

- (a) P fixed  $\longrightarrow P$
- (b) MS = MD  $\longrightarrow R$
- (c) RE  $\longrightarrow E^e$
- (d) UIP  $\longrightarrow E$

## Step 1: Solve initial LR equilibrium ( $t = t_0$ )

(a) Flex P / PPP  $\rightarrow E$  :  $E_0 = kM_0^s = 1 \cdot 1 = 1$

(b) Defn LR / RE  $\rightarrow E^e$  :  $E_0^e = E_0 = 1$

(c) UIP  $\rightarrow R$  :  $R_0 = R^* + \frac{E_0^e}{E_0} - 1$   
 $= 0.08 + \frac{1}{1} - 1 = 0.08$



## Step 1: Solve initial LR equilibrium ( $t = t_0$ )

$$\begin{aligned} \text{(d) } MS=MD \rightarrow P & \quad : \quad \frac{M_0^s}{P_0} = \frac{Y_0}{1 + R_0} \\ & \Rightarrow P_0 = M_0^s \cdot \frac{1 + R_0}{Y_0} \\ & \quad = 1 \times 1.08 = 1.08 \end{aligned}$$

Solution for  $t_0$ :  $P_0 = 1.08$ ,  $R_0 = 0.08$ ,  $E_0^e = 1$ ,  $E_0 = 1$

## Step 2: Solve LR equilibrium ( $t = LR$ )

(a) Flex P / PPP  $\rightarrow E$  :  $E_{LR} = kM_{LR}^s = 1 \cdot 1.05 = 1.05$

(b) Defn LR / RE  $\rightarrow E^e$  :  $E_{LR}^e = E_{LR} = 1.05$

(c) UIP  $\rightarrow R$  :  $R_{LR} = R_{LR}^* + \frac{E_{LR}^e}{E_{LR}} - 1$   
 $= 0.08 + \frac{1.05}{1.05} - 1 = 0.08$

## Step 2: Solve LR equilibrium ( $t = LR$ )

$$\begin{aligned} \text{(d) } MS=MD \rightarrow P & \quad : \quad \frac{M_{LR}^s}{P_{LR}} = \frac{Y_{LR}}{1 + R_{LR}} \\ & \Rightarrow P_{LR} = M_{LR}^s \cdot \frac{1 + R_{LR}}{Y_{LR}} \\ & = 1.05 \times 1.08 = 1.134 \end{aligned}$$

Solution for LR:  $P_{LR} = 1.134$ ,  $R_{LR} = 0.08$ ,  $E_{LR}^e = 1.05$ ,  $E_{LR} = 1.05$

### Step 3: Solve SR equilibrium ( $t = SR$ )

$$(a) \quad P \text{ fixed} \rightarrow P \quad : \quad P_{SR} = P_0 = 1.08$$

$$(b) \quad MS=MD \rightarrow R \quad : \quad \frac{M_{SR}^s}{P_{SR}} = \frac{Y_{SR}}{1 + R_{SR}}$$

$$\Rightarrow \frac{1.05}{1.08} = \frac{1}{1 + R_{SR}}$$

$$\Rightarrow 1 + R_{SR} = \frac{1.08}{1.05}$$

$$\Rightarrow R_{SR} = \frac{1.08}{1.05} - 1 \approx 0.02857$$

### Step 3: Solve SR equilibrium ( $t = SR$ )

(c)  $RE \rightarrow E^e$  :  $E_{SR}^e = E_{LR} = 1.05$

(d)  $UIP \rightarrow E$  :  $R_{SR} = R_{SR}^* + \frac{E_{SR}^e}{E_{SR}} - 1$

$$\Rightarrow 0.02857 = 0.08 + \frac{1.05}{E_{SR}} - 1$$

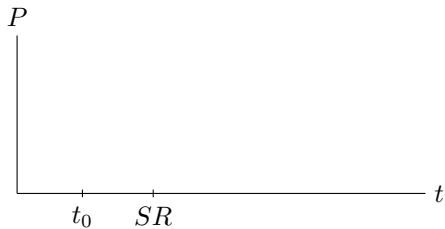
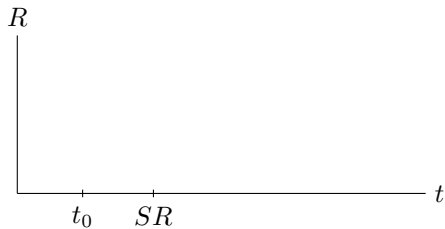
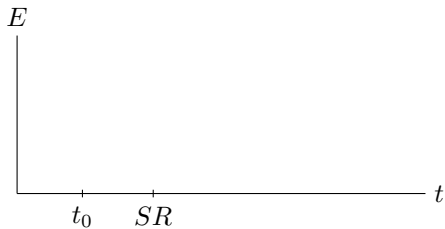
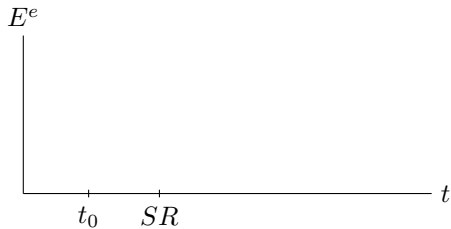
$$\Rightarrow E_{SR} = \frac{1.05}{0.02857 - 0.08 + 1} = \frac{1.05}{0.94857} \approx 1.1069$$

Solution for SR:  $P_{SR} = 1.08$ ,  $R_{SR} \approx 0.029$ ,  $E_{SR}^e = 1.05$ ,  $E_{SR} \approx 1.107$

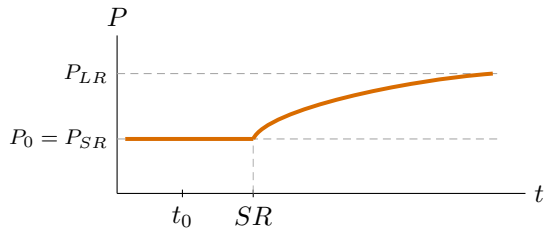
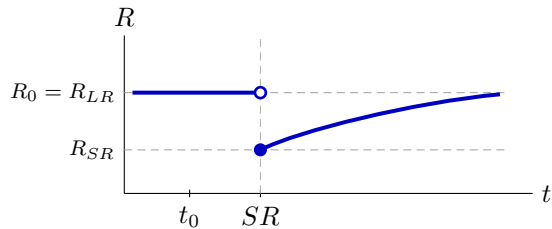
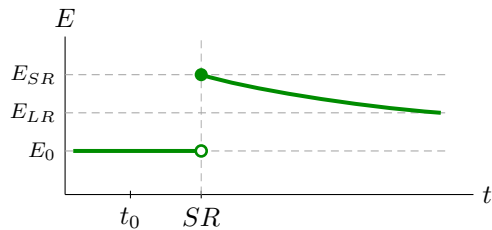
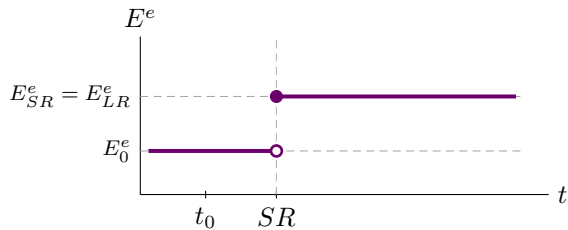
# Example: solution summary

|       | $t_0$ | $SR$  | $LR$  |
|-------|-------|-------|-------|
| $P$   | 1.08  | 1.08  | 1.134 |
| $R$   | 0.08  | 0.029 | 0.08  |
| $E^e$ | 1     | 1.05  | 1.05  |
| $E$   | 1     | 1.107 | 1.05  |

# Paths of solution

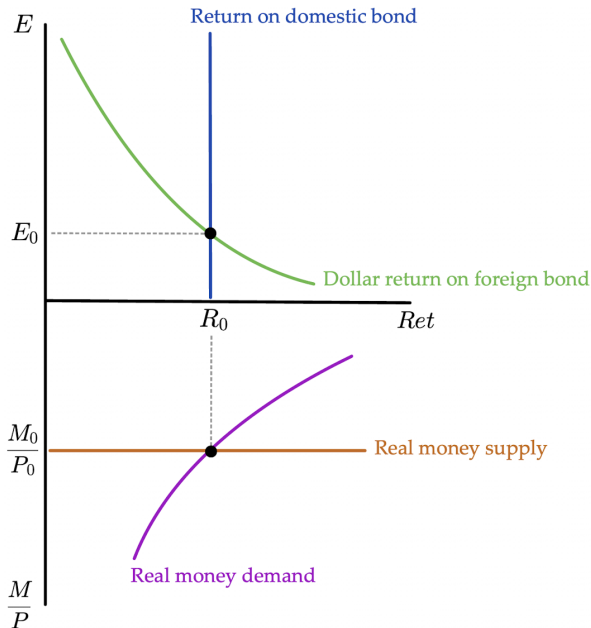


# Paths of solution

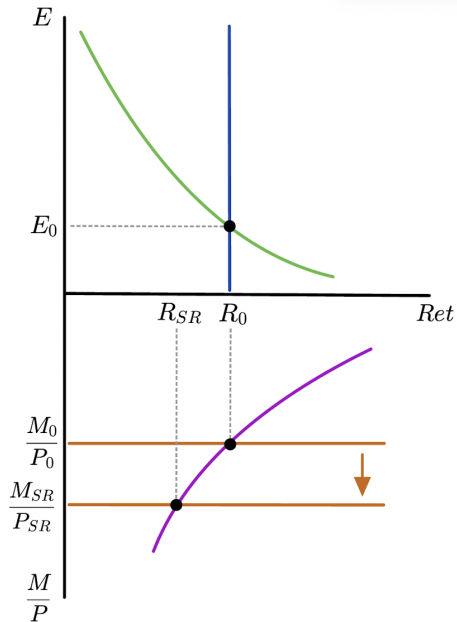




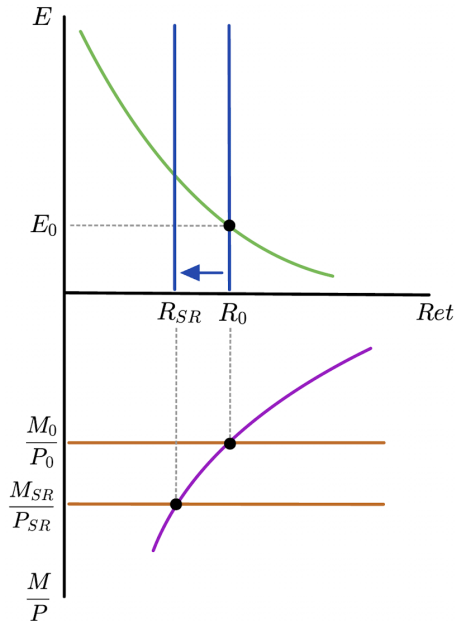
Solve with plots:  
Initial LR  
equilibrium



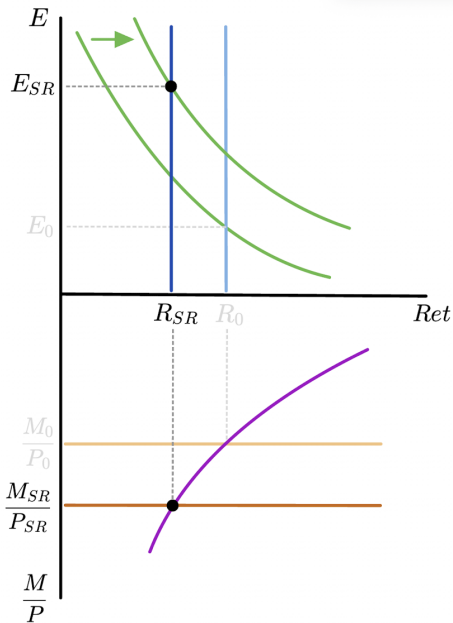
Higher  $M^s$   
lowers  $R$



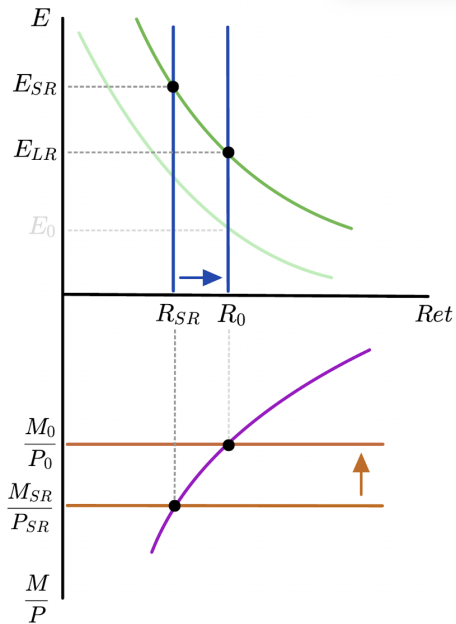
Lower  $R$   
causes  
depreciation



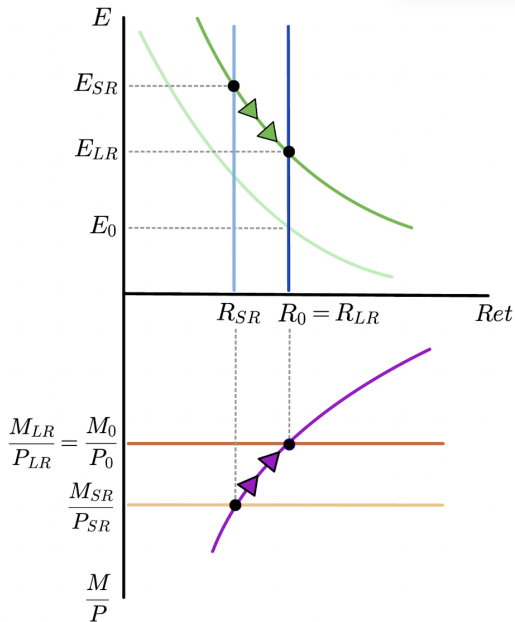
Higher  $E^e$   
shifts dollar  
return on  
foreign bond



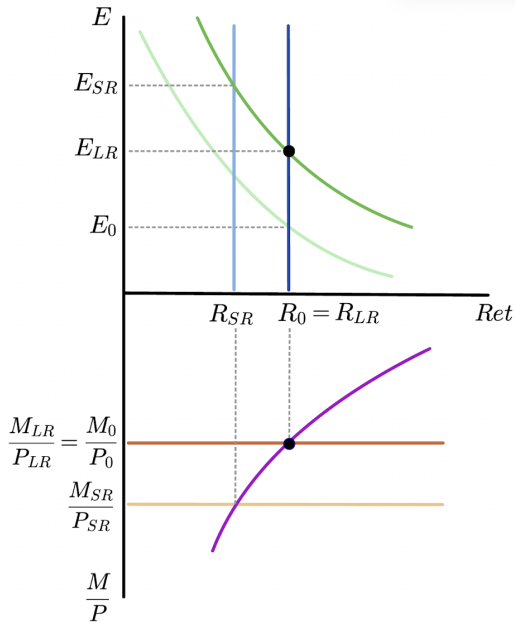
# SR equilibrium



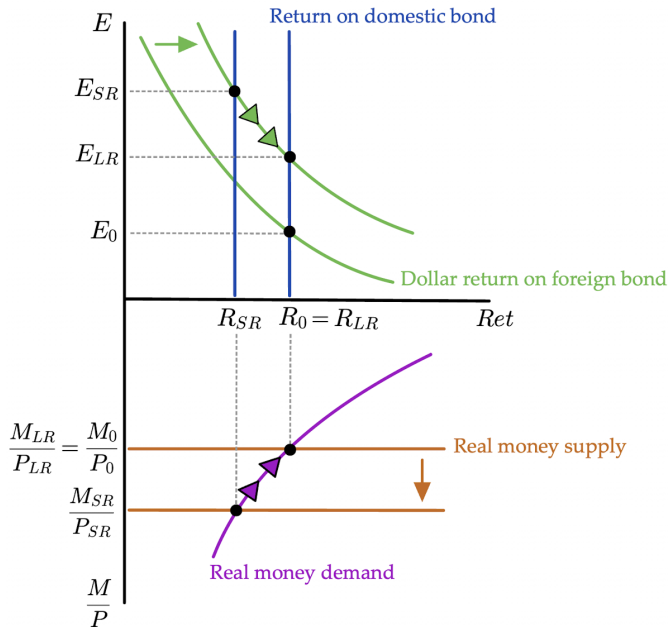
# Transition between SR and LR



Final LR  
equilibrium



Solve with plots:  
All steps  
together





## Equilibrium Determination Map

|       | Initial      | SR        | LR           |
|-------|--------------|-----------|--------------|
| $E$   | flex P / PPP | FX mkt    | flex P / PPP |
| $E^e$ | defn LR      | RE        | defn LR      |
| $R$   | FX mkt       | money mkt | FX mkt       |
| $P$   | money mkt    | fixed P   | money mkt    |

PPP = purchasing power parity  $\Rightarrow E \propto P \propto M^s$

defn LR = definition of the long run  $\Rightarrow E_0^e = E_0$  and  $E_{LR}^e = E_{LR}$

RE = rational expectations  $\Rightarrow \begin{cases} E_{SR}^e = E_{LR} \\ E_0^e = E_0 \text{ and } E_{LR}^e = E_{LR} \end{cases}$

ECON 1550 International Finance

# Price Levels and the Exchange Rate in the Long Run

# Law of One Price (LOOP)

- The LOOP holds if for every good  $i$  with dollar price  $P_{US}^i$  and euro price  $P_E^i$ , we have

$$P_{US}^i = E_{\$/\epsilon} \times P_E^i$$

- This is a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}^i}{P_E^i}$$

# Purchasing Power Parity (PPP)

- PPP holds if

$$P_{US} = E_{\$/\epsilon} \times P_E$$

where  $P_{US}$  is the US price level and  $P_E$  is the euro area price level

- This is also a theory of exchange rate determination:

$$E_{\$/\epsilon} = \frac{P_{US}}{P_E}$$

# Relative PPP

- Relative PPP holds if

$$\frac{E_{\$/\epsilon,t} - E_{\$/\epsilon,t-1}}{E_{\$/\epsilon,t-1}} = \pi_{US,t} - \pi_{E,t}$$

where  $\pi_t$  is inflation  $\pi_t = P_t/P_{t-1} - 1$

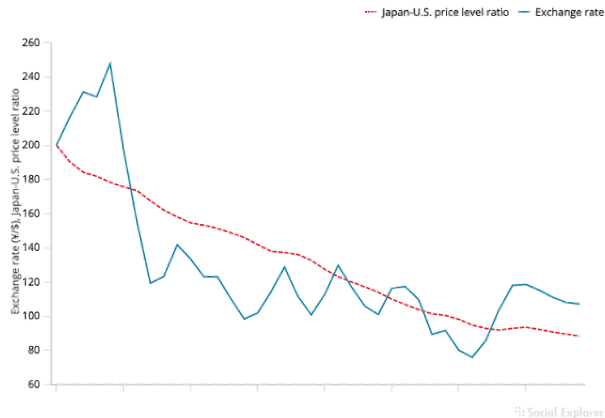
- This is also a theory of exchange rate determination

$$E_{\$/\epsilon,t} = (\pi_{US,t} - \pi_{E,t} + 1)E_{\$/\epsilon,t-1}$$

Relative PPP  
does not hold  
very well in  
the real world

Figure 16-2

The Yen/Dollar Exchange Rate and Relative Japan-U.S. Price Levels, 1980–2019



The graph shows that relative PPP does not track the yen/dollar exchange rate during 1980–2015.

**Source:** IMF, *International Financial Statistics*. Exchange rates and price levels are end-of-year data.

# Problems With PPP

- Price levels of different countries report different baskets of goods
  - E.g. GDP deflator is the basket of goods in GDP for each country
- Deviations from perfectly competitive frictionless markets
  - Transport costs and trade barriers
  - Monopoly power

# Expected Relative PPP

- Expected relative PPP (or relative PPP in expectations)

$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e$$

where  $\pi^e$  is *expected* inflation  $\pi^e = P^e/P - 1$

- This is also a theory of exchange rate determination

$$E_{\$/\epsilon} = \frac{E_{\$/\epsilon}^e}{\pi_{US}^e - \pi_E^e + 1}$$



# Relationships

- LOOP  $\Rightarrow$  PPP  $\Rightarrow$  Relative PPP  $\Rightarrow$  Expected relative PPP
- Expected relative PPP + UIP  $\Rightarrow$  Fisher effect

$$\frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}} = \pi_{US}^e - \pi_E^e \quad \text{and} \quad R_{\$} = R_{\epsilon} + \frac{E_{\$/\epsilon}^e - E_{\$/\epsilon}}{E_{\$/\epsilon}}$$

imply

$$R_{\$} - R_{\epsilon} = \pi_{US}^e - \pi_E^e$$

# Common variables across all models

## Exogenous variables

| Variable       | Description               |
|----------------|---------------------------|
| $R^*$          | Foreign interest rate     |
| $M^s, M^{s*}$  | Money supply              |
| $g_M, g_{M^*}$ | Money supply growth rates |

## Endogenous variables

| Variable      | Description         | Equation                   | Type of equation           |
|---------------|---------------------|----------------------------|----------------------------|
| $P$           | Price level         | $M^s/P = L(R, Y)$          | Behavioral + eq. condition |
| $P^*$         | Foreign price level | $M^{s*}/P^* = L(R^*, Y^*)$ | Behavioral + eq. condition |
| $\pi, \pi^*$  | Inflation           | $\pi = g_M - g_L$          | Definition                 |
| $r^e, r^{e*}$ | Real interest rates | $r^e = R - \pi^e$          | Definition                 |

# Model 1: PPP

## Exogenous variables

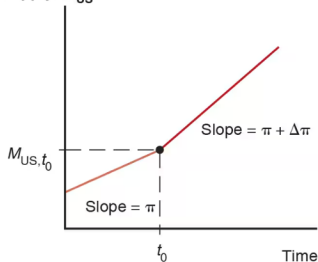
| Variable | Description            |
|----------|------------------------|
| $R$      | Domestic interest rate |
| $Y, Y^*$ | Real incomes           |

## Endogenous variables

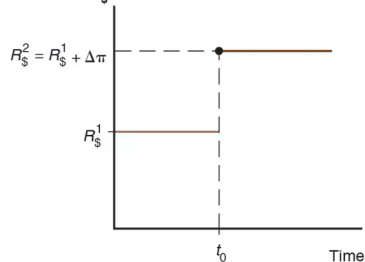
| Variable | Description   | Equation    | Type of equation |
|----------|---------------|-------------|------------------|
| $E$      | Exchange rate | $E = P/P^*$ | Behavioral       |

# Changes in growth rate of money supply

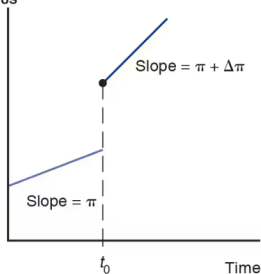
(a) U.S. money supply,  $M_{US}$



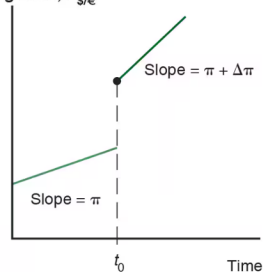
(b) Dollar interest rate,  $R_{\$}$



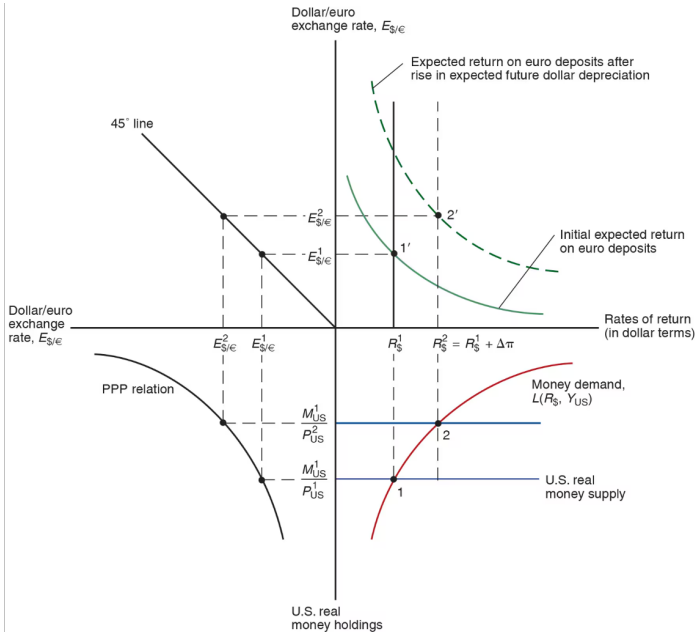
(c) U.S. price level,  $P_{US}$



(d) Dollar/euro exchange rate,  $E_{\$/\epsilon}$



# Changes in growth rate of money supply



# Model 2: PPP<sup>e</sup> + UIP

| Exogenous variables |              | Endogenous variables |                       |                                |                  |
|---------------------|--------------|----------------------|-----------------------|--------------------------------|------------------|
| Variable            | Description  | Variable             | Description           | Equation                       | Type of equation |
| $Y, Y^*$            | Real incomes | $E^e/E - 1$          | Expected depreciation | $E^e/E - 1 = \pi^e - \pi^{e*}$ | Behavioral       |
|                     |              | $R$                  | Interest rate         | $R = R^* + E^e/E - 1$          | Eq. condition    |

## Model 3: real $E$ + relative output

- The real exchange rate

## Model 3: real $E$ + relative output

- Real exchange rate calculation



# Model 3: real $E$ + relative output

## Exogenous variables

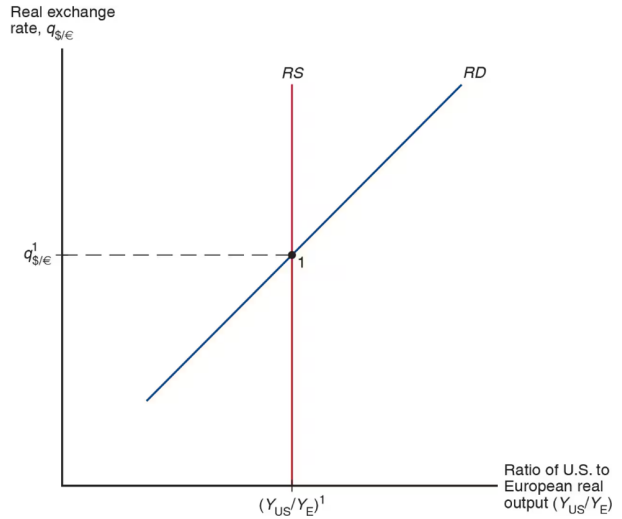
| Variable | Description            |
|----------|------------------------|
| $RS$     | Relative output supply |
| $q$      | Real exchange rate     |

## Endogenous variables

| Variable | Description           | Equation     | Type of equation |
|----------|-----------------------|--------------|------------------|
| $Y/Y^*$  | Relative output       | $RD(q) = RS$ | Eq. cond.        |
| $E$      | Nominal exchange rate | $E = qP/P^*$ | Definition       |

# Model 3 equilibrium

## Determination of the Long-Run Real Exchange Rate



ECON 1550 International Finance

# Output and the Exchange Rate in the Short Run

# Behavioral equations in the goods market

Demand for domestic goods

# Equilibrium in the goods market

# A short-run model of the goods market

## Exogenous variables

| Variable | Description           |
|----------|-----------------------|
| $E$      | Nominal exchange rate |
| $I$      | Investment            |
| $G$      | Government spending   |
| $T$      | Taxes                 |
| $P$      | Price level           |
| $P^*$    | Foreign price level   |
| $Y^*$    | Foreign income        |

### Endogenous variables

| Variable | Description               | Equation                                             | Type of equation      |
|----------|---------------------------|------------------------------------------------------|-----------------------|
| $Y$      | Income, production        | $Y = D$                                              | Equilibrium condition |
| $Y_D$    | Disposable income         | $Y_D \equiv Y - T$                                   | Identity              |
| EX       | Exports                   | $EX = EX(q, Y^*)$<br>(+) (+)                         | Behavioral            |
| IM       | Imports                   | $IM = IM(q, Y_D)$<br>(-) (+)                         | Behavioral            |
| CA       | Current account           | $CA \equiv EX - IM = CA(q, Y_D, Y^*)$<br>(+) (-) (+) | Identity              |
| $D$      | Demand for domestic goods | $D \equiv C + I + G + CA$                            | Identity              |
| $A$      | Domestic demand           | $A \equiv C + I + G$                                 | Identity              |
| $C$      | Consumption               | $C = C(Y_D)$<br>(+)                                  | Behavioral            |
| $q$      | Real exchange rate        | $q \equiv \frac{EP^*}{P}$                            | Identity              |



# DD Curve

$$Y = D$$

$$Y = C + I + G + CA$$

$$Y = \underset{(+)}{C}(Y_D) + I + G + \underset{(+)}{CA}(\underset{(-)}{q}, \underset{(+)}{Y_D}, Y^*)$$

$$Y = \underset{(+)}{C}(Y_D) + I + G + \underset{(+)}{CA}(\underset{(-)}{EP^*/P}, \underset{(+)}{Y_D}, Y^*)$$

# Short-run FX and money market model

## Exogenous variables

| Variable | Description            |
|----------|------------------------|
| $R^*$    | Foreign interest rate  |
| $E^e$    | Expected exchange rate |
| $Y$      | Real income            |
| $M^s$    | Money supply           |
| $P$      | Price level            |

## Endogenous variables

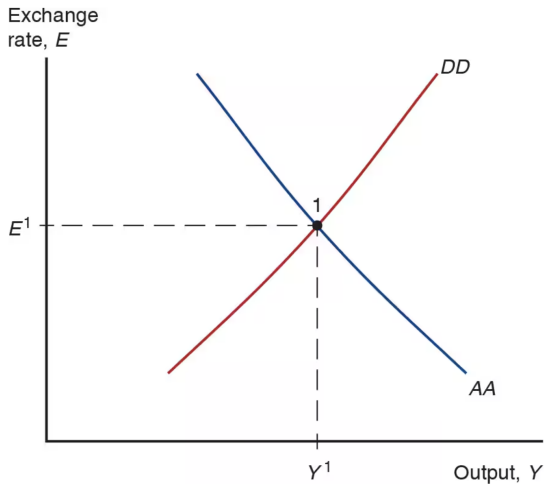
| Variable | Description            | Equation                      | Type of equation      |
|----------|------------------------|-------------------------------|-----------------------|
| $E$      | Exchange rate          | $R = R^* + \frac{E^e - E}{E}$ | Equilibrium condition |
| $R$      | Domestic interest rate | $M^d/P = L(R, Y)$             | Behavioral equation   |
| $M^d$    | Money demand           | $M^d = M^s$                   | Equilibrium condition |

# AA Curve

$$\text{(UIP):} \quad R = R^* + \frac{E^e}{E} - 1$$

$$\text{(MS) = (MD):} \quad \frac{M^s}{P} = L(R, Y)$$

# Short-run Equilibrium in $AA-DD$ model



# Example: DD Schedule

$$C_{(+)}(Y_D) = 1 + 0.75Y_D$$

How to find the DD schedule

$$EX_{(+)}(q, Y^*)_{(+)} = 0.15 + 0.5q + 0.1Y^*$$

$$IM_{(-)}(q, Y_D)_{(+)} = 0.1 - 0.2q + 0.12Y_D$$

$$Y_D = Y - T$$

$$q = \frac{EP^*}{P}$$

# Example: AA Schedule

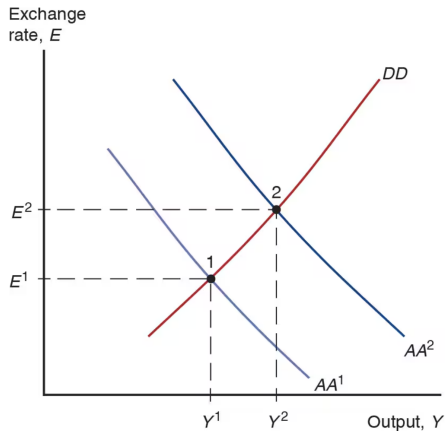
$$R = R^* + \frac{E^e - E}{E}$$

How to find the AA schedule

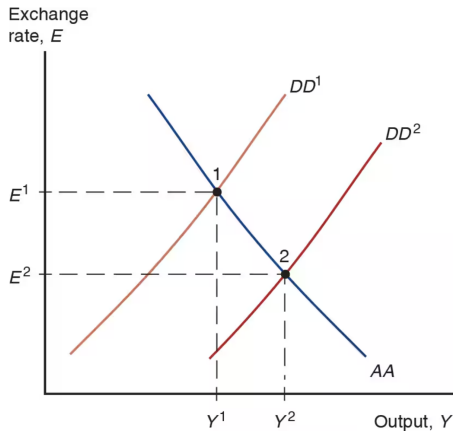
$$\frac{M^d}{P} = \frac{M^s}{P}$$

$$\begin{aligned}\frac{M^d}{P} &= L_{(-)}(R, Y_{(+)}) \\ &= 1.1 - R + 0.05Y\end{aligned}$$

# Temporary Change in Monetary Policy



# Temporary Change in Fiscal Policy





# Full AA-DD Model

DD Schedule:  $Y = C(Y_D) + I + G + CA(EP^*/P, Y_D, Y^*)$

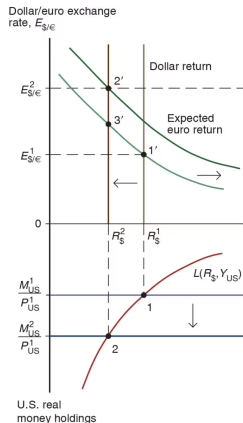
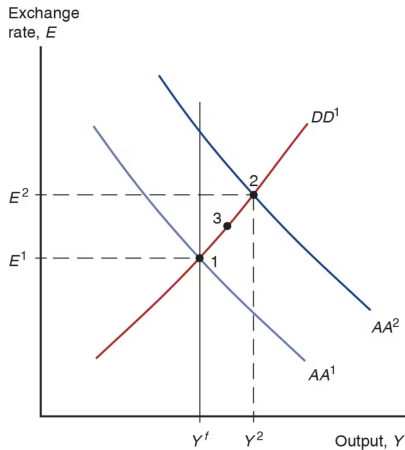
AA Schedule:  $\frac{M^s}{P} = L\left(R^* + \frac{E^e}{E} - 1, Y\right)$

Phillips Curve:  $\pi = \pi^e + \alpha(Y - Y^f)$

Definition of inflation:  $\pi_t = \frac{P_t}{P_{t-1}} - 1$

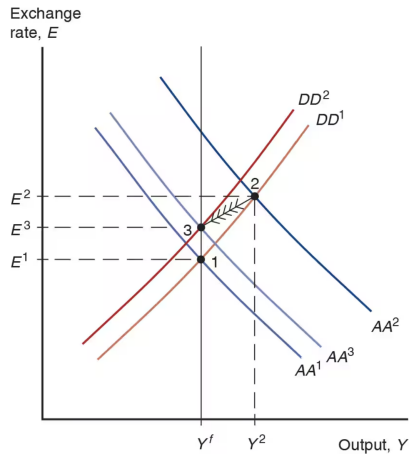
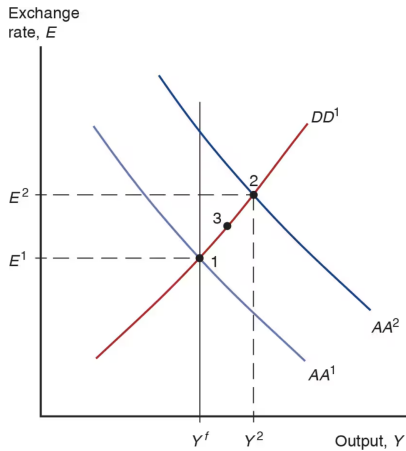
Definition of expected inflation:  $\pi_t^e = \frac{P_{t+1}^e}{P_t} - 1$

# Permanent Shifts in Monetary Policy

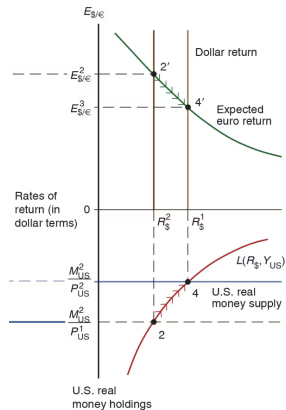
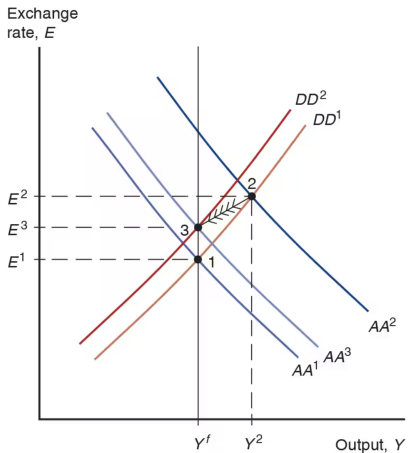


(a) Short-run effects

# Permanent Shifts in Monetary Policy

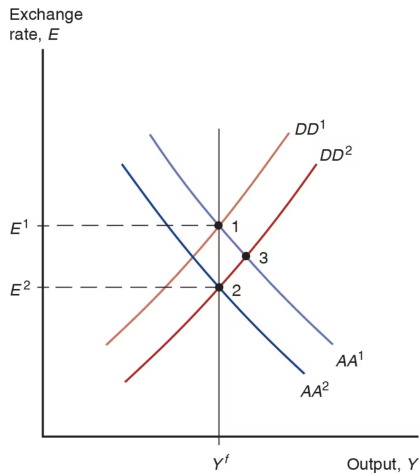


# Permanent Shifts in Monetary Policy



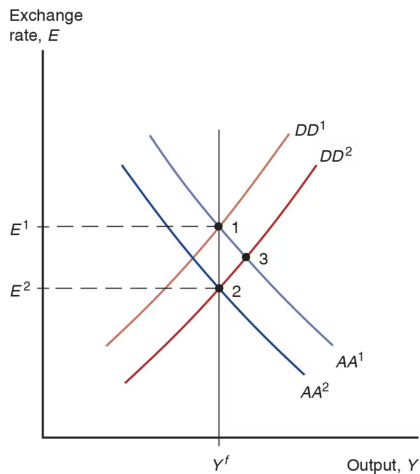
(b) Adjustment to long-run equilibrium

# Permanent Shifts in Fiscal Policy



- Start in a medium run equilibrium at point 1 with  $R_0 = R^*$  and  $E_0^e = E^1$
- Increase in  $G$  shifts  $DD$  to the right
- Since the shift in  $DD$  is permanent,
- At point 3, exchange rate is lower than  $E_1$
- appreciation at point 3 is also expected to be permanent at point 3
- $E^e$  must also go down
- Lower  $E^e$  shifts  $AA$  down
- But how much?

# Permanent Shifts in Fiscal Policy



- There are no more shifts expected to occur,  $E^e$  remains constant
- Therefore  $E = E^e$
- Real money supply is constant
- Real money demand must stay unchanged
- UIP implies  $R$  does not change
- $Y$  must be unchanged as well to keep real money demand unchanged
- If  $Y$  must be equal to its original level  $Y^f$ ,  $AA^1$  must shift to  $AA^2$

# DD Schedule with Tariffs

# Slope of DD Curve with Tariffs